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Laser-Induced Microfabrication of Carbon Nanostructure: Processing Mechanism and Application for Next-Generation Battery Technology

*Sujit Deshmukh, Shaista Nouseen, and Martin Pumera**

Direct laser writing with a single beam of multiple parallel optical configurations facilitate the production of conductive carbon-based nanomaterials, carbon-based catalysts, micropatterned carbon surface, hierarchical 3D porous carbon structure, numerous carbon-based composites, and so on. This chemical-free, cost-effective, binder-free, maskless patterning technology minimizes the environmental impact and enables more sustainable production of diverse carbon-based materials with a wide range of applications. In this review, 1st a comprehensive overview of the state-of-the-art advancements and formation mechanism of laser processed carbon (LPC) is provided from various carbon-rich precursors (polyimide/PI, polytetrafluoroethylene, poly (ether sulfone)), as well as natural resources like wood, lignin, clothes, paper, and even food are explored. Additionally, the article reports a comprehensive overview of how different laser types, processing conditions, and environmental factors influence the resulting structure and surface chemistry of LPC. Next, specifically, the emerging applications of these LPC in battery technology are focused on. This includes the carbon anode of Li-ion/Na-ion battery, the current collector (CC) for Li-metal battery, electrode for Li-sulfur battery, the catalyst for Li-air batteries, and electrodes for Zn-ion and Zn-air battery. The article concludes with the insights and future perspectives on the advancement of this processing technology for next-generation smart and sustainable battery materials.

1. Introduction

Graphene, a 2D versatile carbon material and its derivatives, have emerged as a viable alternative to a broad range of device applications due to their exceptional electrical, thermal, optical, and mechanical performance.[1–5] Huge efforts have been endowed in the past 10 years to accelerate the expansion of graphene-based flexible electronic devices, ranging from electronic skin to wearable energy storage devices.[6–9] Numerous methodologies have been used to synthesize high-crystalline graphene, which include mechanical/chemical exfoliation,[10,11] chemical vapor deposition (CVD),[12,13] epitaxial growth,[14,15] etc. Another class of methods includes graphene oxide (GO) reduction via thermal, electrical, microwave, and photo treatment.[16–20] Additionally, in many applications, graphene was engineered into a three-dimensional (3D) architecture by either assembling GO into foam or CVD deposition on a 3D porous substrate.[21–23] Most of the conventional methods, however, introduce unwanted contamination through a complex synthesis route or suffer from the need of high scale production of GO through oxidative synthesis routes or hinder the scalable production using high-temperature CVD deposition and subsequent etching and drying process. Simple and effective technologies for graphene preparation are therefore still in high demand to ease practical application.

The laser-assisted reduction process is very promising and widely employed in reducing GO into graphitic materials.[24–26] Interestingly, the laser processing technology

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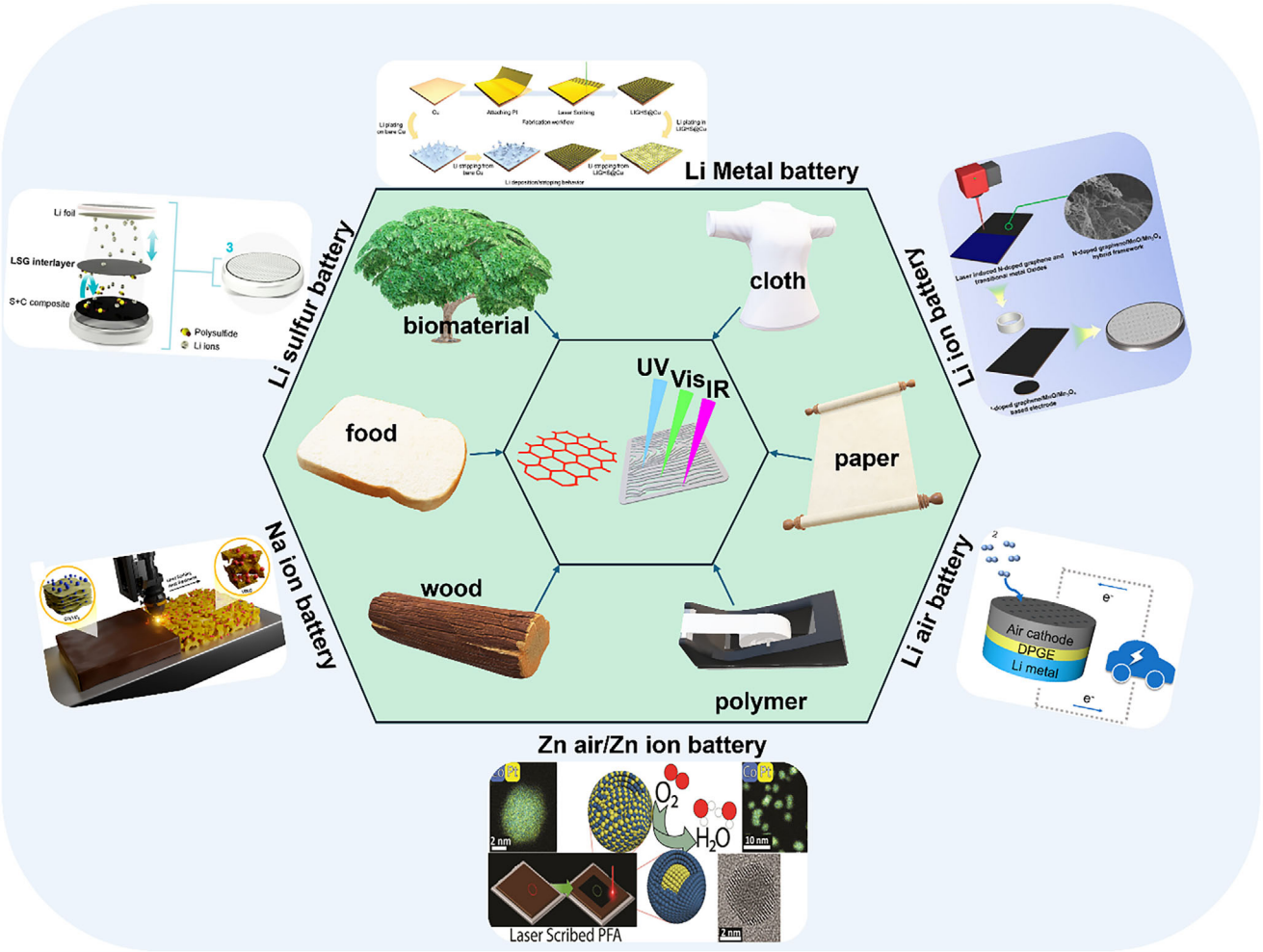


Figure 1. Fabrication schematic of laser-processed carbon using various lasers and different carbon precursors. Direct laser writing/patterning enables simple microfabrication and flexible design of LPC patterns with required physical, chemical properties, which can be utilized effectively in advanced anode/ cathode or CCs in Li-metal batteries, Li ion/sulfur/air batteries, Na-ion batteries, Zn-ion/air batteries. Reproduced (Adapted) with permission.[39,42–46] Copyright 2019, Wiley-VCH; Copyright 2020, American Chemical Society; Copyright 2020, American Chemical Society; Copyright 2019, Elsevier; Copyright 2023, American Chemical Society; Copyright 2024, American Chemical Society.

is directly converting synthetic polymers like polyimide,[27–29] polytetrafluoroethylene,[30] poly(ether sulfone)[31,32] as well as natural materials like wood,[33,34] lignin,[35,36] cloths, paper, and even food[37] into 3D porous graphene structure. The laser-assisted approach eliminates the need for sophisticated synthesis methods and a high-temperature environment. A CO_2 laser (10.6 µm) can directly convert the sp^3 carbon into sp^2 carbon through a photothermal effect.[28] The laser processing technology enables flexible patterning without masks,[38,39] hierarchical micro/nanostructures,[26] and controllable tailoring of graphene properties by heteroatom doping.[40,41] On flexible polymer substrates, any desired graphene pattern can be generated, simplifying the procedures for flexible graphene devices. The whole process can be done within minutes which greatly accelerate the production efficiency of graphene. The resulting laser-induced graphene (LIG) /LPC exhibits high surface area, thermal stability, and conductivity in the range of ≈340 $m^2\ g^{-1}$, >900 °C, and 5–25 S cm^{-1}, respectively.[28] These distinct benefits of laser technology enable the LIG materials being used in a diversity of fields, involving sensors, supercapacitors, batteries, energy generators, optoelectronics devices, water treatment, and so on.

In this report, we comprehensively review the recent development of LPC and its derivatives toward the advancement of flexible electronics. The laser processing of GO and flexible polymers using different types of laser sources is reviewed in **Figure 1**. The features, functionalities of the laser-assisted graphene and their synthesis approaches using GO, and different types of polymers are elucidated in-depth and analyzed. The development of graphene-based flexible electronics has been considerably aided by the laser processing of graphene in the last five years. The LPC-based next generation baterries has been comprehensively reviewed. Finally, the challenges associated with LPC-based battery devices are discussed from the current perspective with possible solution listed.

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2. Precursors for Laser-Processed Carbon Fabrication

2.1. Graphene Oxide as a Precursor

As a conventional precursor for LPC, GO consists of several carbon atom layers that resemble the structure of graphene, but the planes of GO are attached with oxygen functionality groups. As a result, this structural deficiency made GO an insulator. However, the oxygen functionalities make GO hydrophilic, which is beneficial for forming stable aqueous colloid. The common approach to get reduced graphene's structure with exceptional functionalities, the oxygen atoms in GO need to be removed via reduction.

The viability and utility of laser reduction of GO have been established in the literature, although most of these studies were conducted in either an inert gas or vacuum environment.[47–51] Reducing GO to graphene through photoreduction under ambient conditions presents the most scalable method for graphene production.[26,52] With the laser writing approach, GO can be reduced effectively under ambient conditions.[26] **Figure 2a** illustrates that the GO film was exposed with infrared (IR) beam of laser inside a light-scribed optical drive (CD/DVD). In this process, GO undergoes photothermal conversion, where it absorbs high-intensity laser energy and transforms into graphene. By using laser accuracy, graphene can be patterned into interdigitated circuits as well.[53] A clear visible change in color from brown to black was witnessed and well exfoliated LPC was obtained through the laser process. The resulting LPC showed exceptional electrical conductivity (≈1738 S m^{-1}) and mechanical flexibility (≈1% variation in the film's electrical resistance was observed after 1000 bending cycles). Very recently, Fatkullin, et al. illustrate the LPC fabrication schematic using three laser wavelengths: 405, 532, and 633 nm and different laser powers (Figure 2b).[54] Alternative strategy to improve electric conductivity is to mix GO with metal salts (like gold, cobalt, etc.) or with the nitrogen-based element (GaN) (Figure 2c,d).[55–57] Metal salts were converted to metal and deposited on the surface of LPC during laser irradiation, while GaN acted as a nitrogen dopant to the LPC surface.

The reduction, however, depends on the laser parameters and types of lasers being used for graphene formation. Petridis et al. reported that a threshold fluence of 10-15 mJ cm^{-2} was necessary to begin the elimination of groups contain oxygen from the GO lattice, while exceeding the fluence above 80 mJ cm^{-2} can degrade or damage the GO by partial ablation of the GO layers.[51] Pulse temporal width and laser energy distribution are also influencing factors for the photoreduction of GO. Arul et al. showed that nanosecond UV (248 nm) pulses were the most effective to produce graphene-like films as compared to 788 nm continuous wave illumination and 100 fs. near-infrared (800 nm) pulses.[52]

2.2. Polyimide Sheet as a Precursor

In the early 1990s, the PI film was exposed with a UV laser to produce electrically conductive porous carbon materials.[58,59] However, the significant method to produce graphene from PI sheets is introduced by Tour et.al in the year 2014.[28] There, they used a CO_2 IR laser to convert a commercially available PI film into porous LIG under ambient environment. The high-quality graphene formation is validated by the D/G intensity ratio obtained from Raman spectroscopy analysis where it displays three distinct peaks: the D peak around 1350 cm^{-1} (indicative of defect-induced sp^2 carbon), the G peak near 1,580 cm^{-1} (corresponding to the first-order allowed vibrational mode), and the 2D peak approximately at 2700 cm^{-1}. Here, the graphene layers are packed along the C axis, and LIG exhibits high electrical conductivity ≈25 S cm^{-1}. Important to note that, LPC can be easily transferred to a stretchable transparent substrate such as PDMS (polydimethylsiloxane) and silicon rubber.[60,61] Depending on the application, we can now stretch the device to the length of 400% of its original dimension. To further expand the capability of LIG, manganese dioxide (MnO_2), ferric oxyhydroxide (FeOOH), and polymers like polyaniline were electrodeposited onto its surface.[62] To improve the electrical and electrochemical performance, another group reported the boron and fluorine Co-doped LPC, which has a high energy storage capacity as compared to bare LIG.[63,40] Here they synthesized PI sheets using fluorinated polyamic acid or F-PAA solution, and H_3BO_3 was used as the source of boron. The LIG morphology and surface chemical oxidation states can also be altered by modulating the laser surrounding atmosphere. The water contact angle ranged from 0° (indicating superhydrophilicity) under an air or O_2 atmosphere to over 150° (indicating superhydrophobicity) when processed in an Ar or H_2 atmosphere.[64] The effect of laser fluence (regardless of the laser power) also plays a substantial role in controlling the morphology. It was determined that a threshold fluence of ≈5 J cm^{-2} is needed to initiate the carbonization process of PI film. It's worth noting that one-dimensional (1D) LIG fibers can be made by a one-step laser photo thermolysis method at radiation fluence of >40 J cm^{-2}.[65] These LIG fibers are superior in terms of energy storage performance compared to 3D LIG. Later, based on the different thicknesses of the PI sheet, Carvalho et al. and Santos et al. studied the effect of the laser source.[66,67] They demonstrated that LIG can be prepared from PI sheets using an ultraviolet (UV) laser rather than a conventional infrared laser. Because, UV laser has a greater scribing resolution than IR laser, it is possible to manufacture miniaturized devices. Very recently, Rehman et al. used 60-watt laser engraver (10.6 µm light source) to transform the PI film into LPC. They prepared a unique porous free standing LPC anode for Li-ion battery (LIB) (Figure 3a).[68] In another work Hossen et al. reports an single-step and simple lasing route to prepare laser-induced graphite graphene matrix LIB anode from graphite coated PI sheet (Figure 3b).[69] The laser conversion of a PI sheet into a graphene structure is commercially established and compatible with roll-to-roll manufacturing processes.

2.3. Other Polymers and Papers as a Precursor

In the previous section, we have discussed how aromatic, and repeat units of imide in Kapton PI can be converted to a graphene structure using both an IR and UV laser. Apart from the PI sheet, Singh et al. showed the aromatic polysulfone-based polymers such as polysulfone (PSU), polyphenylsulfone (PPSU) and poly(ether sulfone) (PES) can also be transformed into graphene structure using a CO_2 laser in ambient atmosphere (**Figure 4a**).[31] Interesting to note that, LIG layers produced from these class polymers were doped with sulfur as a result of the

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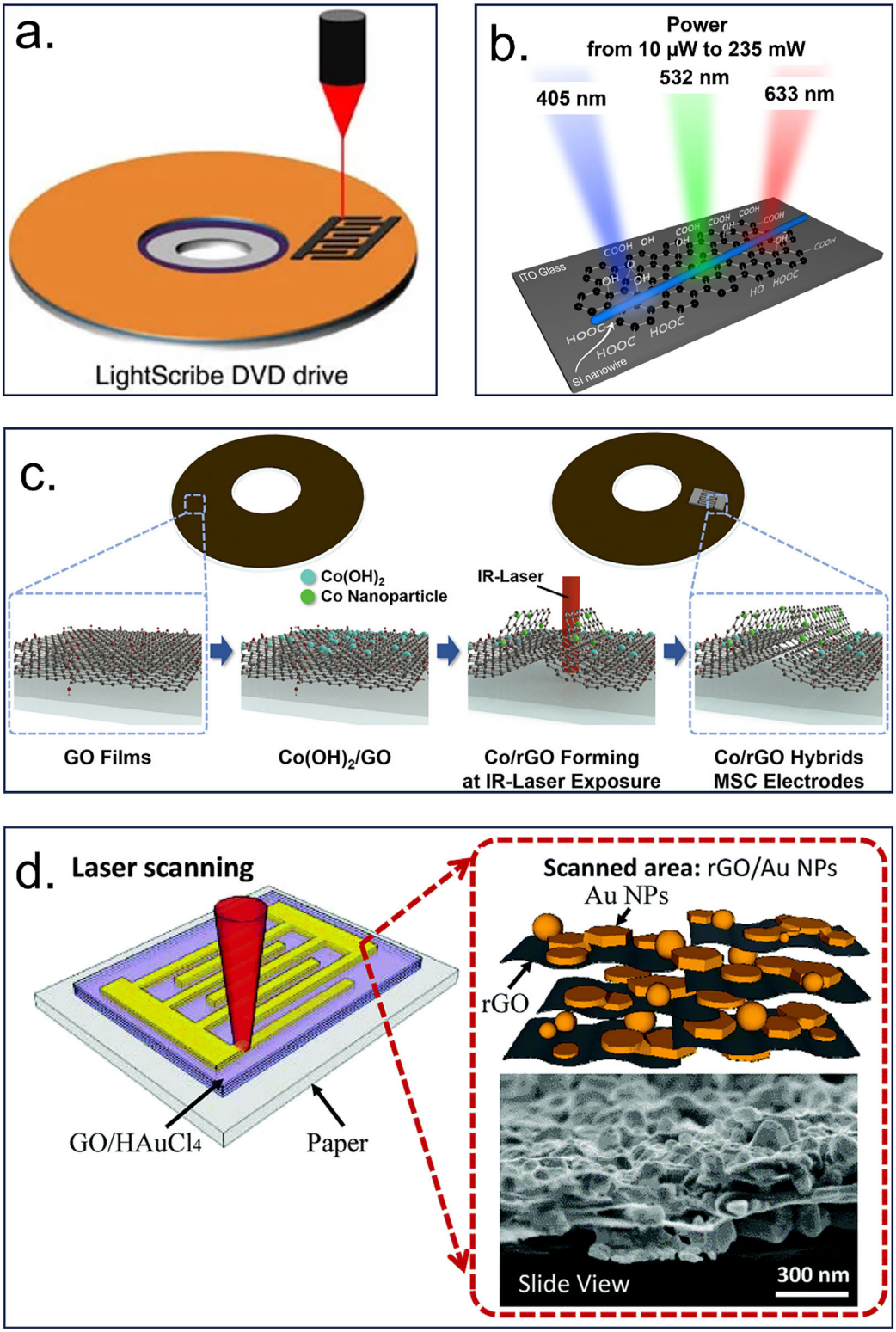


Figure 2. Fabrication schemes of LPC from GO precursors. a) A schematic showing the laser fabrication process for patterned graphene from graphene oxide. Reproduced (Adapted) with permission.[53] Copyright 2013, Springer Nature. b) Laser-reduction of graphene oxide mixed with silicon nanowires on ITO/glass using lasers with various wavelengths and powers. Reproduced (Adapted) with permission.[54] Copyright 2024, Springer Nature. c) A Schematic diagram of the laser manufacture process to produce Co/reduced graphene oxide electrodes. Reproduced (Adapted) with permission.[55] Copyright 2021, American Chemical Society. d) Schematic demonstration for the laser-reduced fabrication of graphene/Au composite. Reproduced (Adapted) with permission.[56] Copyright 2016, Royal Society of Chemistry.

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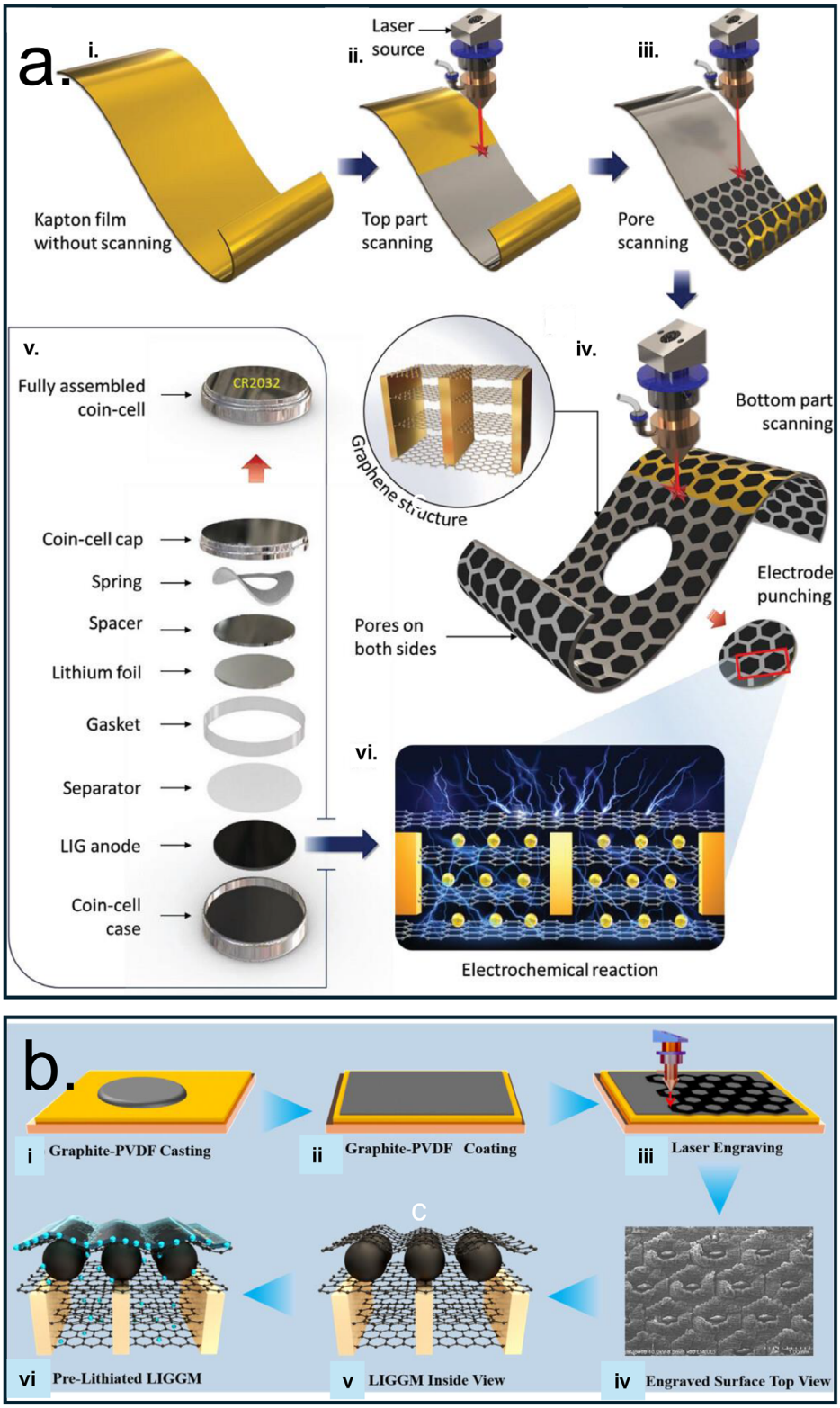


Figure 3. Fabrication schemes of LPC from PI sheets and their battery assembly. a) Schematic illustration of LIG fabrication and coin cell assembly for LIB. i) PI sheet before laser scanning. ii) Laser scanning on top of PI film. iii) Laser scanning on the PI films pore area. iv) Laser scanning on bottom part and punching the PI film. v) coin cell assembly for LIB. vi) Electrochemical reaction on LIG anode. Reproduced (Adapted) with permission.[68] Copyright 2024, Wiley-VCH. b) Schematic demonstration of the fabrication of laser-induced graphite graphene matrix anode and visualization of pre-lithiated state. Reproduced (Adapted) with permission.[69] Copyright 2025, Elsevier.

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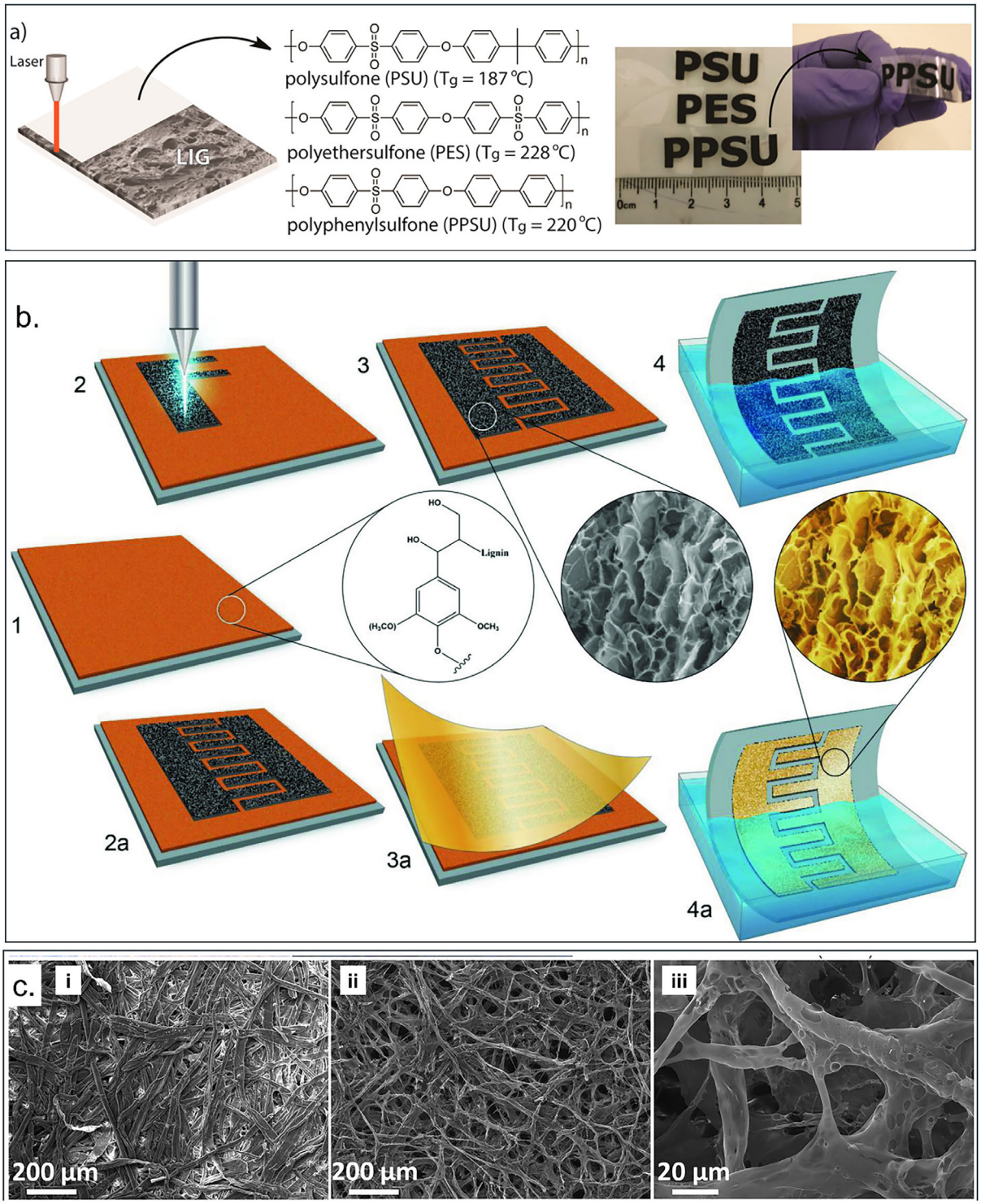


Figure 4. LPC production from various polymers and papers. a) Schematic of polysulfone class polymers used for laser processed graphene fabrication. Reproduced (Adapted) with permission.[31] Copyright 2017, American Chemical Society. b) Schematic diagram showing the lignin-derived graphene preparation using laser lithography process. 1) Deposition of a lignin/PVA film onto a substrate using blade coating; 2) laser rastering to convert lignin into graphene; 3) formation of interdigitated graphene patterns through laser processing; 4) completion of the lithographic process by removing unexposed lignin film via water-based lift-off; 2a) interdigitated graphene patterns created by laser processing before gold deposition; 3a) Gold sputtering onto the laser-formed graphene electrodes; 4a) final water-based lift-off step to remove unexposed lignin after laser treatment. Reproduced (Adapted) with permission.[36] Copyright 2018, Wiley-VCH. c) SEM images display the fire retardant-treated filter paper (i) prior to and (ii) after graphene formation. (iii) Close-up of the microstructure of the paper-based graphene. Reproduced (Adapted) with permission.[72] Copyright 2021, American Chemical Society.

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intrinsic sulfur present in these polymers. These polymers yielded a new class of sulfur doped LIG with excellent electrical, antibacterial, and antibiofilm activity. With a similar method, Sun et al. also prepared sulfur-doped LIG by integrating lignin and PES polymer.[70] Subsequently, Alshareef and colleagues introduced an innovative and straightforward lignin-based laser lithography method (Figure 4b).[36] Lignin a complex phenylpropane-based reticular aromatic polymer, is unsuitable for the traditional chemical engineering processes. But, due to its rich aromatic ring content, lignin is considered a good carbon precursor. A straightforward CO_2 laser irradiation converted lignin films into 3D laser-scribed graphene (LSG) electrodes. The resulting lignin-derived graphene has remarkable electrical conductivity (≈66.2 S cm^{-1}) and high surface area (≈338 m^2 g^{-1}). In the same year, Tour and his team presented a significant method for producing fluorinated graphene by irradiating polytetrafluoroethylene (PTFE or Teflon) with a 9.3 µm pulsed CO_2 laser under an argon atmosphere, eliminating the need for an external fluorine source.[30] Here, the lasing atmosphere was an important parameter, and the production of fluorinated graphene was done with a defocused laser spot to lase the surface multiple times.

Aside from the polymers mentioned above, various types of papers were also used in the laser process to obtain graphene. Chyan et al. showed that patterned graphene could be fabricated directly on various paper substrates such as cotton paper, cardboard boxes, Whatman filter paper, and so on.[37] Park et al. created a similar type of paper-based LIG for use in paper electronics.[71] To limit ablation, these papers were preserved with a commercial phosphate fire retardant and then dried before the lasing process. Later, Kulyk et al. used Cellulose filter paper to synthesize LIG by exposing the paper to the laser two times, 1st at 7 mm beneath the focal plane and again in focus. The corresponding microstructure of paper-based graphene is displayed as Figure 4c.[72] Recently, Pinheiro et al. prepared LIG from chromatographic and office papers. Before the lasing process, these papers were soaked in sodium borate solution, and then sheets were submitted to wax printing. The Raman peaks confirmed the multi-layer graphene formation, and the sheet resistance of this paper-based LIG is also as low as 56.0 Ω sq^{-1}. The popular laser type, operational parameters, and reported applications are summarized in **Table 1**.

2.4. Comparative Discussion of the Relative Advantages and Limitations of Precursors

GO offers ease of reduction and easy patterning with impressive electrical properties but often requires controlled atmosphere (inert gas or vacuum) for high-quality graphene formation and also suffer from the limitations in scalability due to chemical synthesis complexity. On the other hand, commercial PI sheets are a widely used synthetic polymer precursor for laser processing in ambient condition. The conductive carbon produced from PI sheet possesses excellent conductivity (5–25 S cm^{-1}) and high degree of flexibility. PI-based LPC is commercially scalable and compatible with roll-to-roll manufacturing, but the high-quality PI sheet is expensive. Polymers like PES and PSU inherently introduce, sulfur atom doping which improves catalytic properties, but laser processing parameters need careful optimization to preserve these functionalities. Natural carbon precursors such as lignin are aligned well with sustainability goal and low-cost materials, with lignin derived LPC showing outstanding electrical conductivity. Nonetheless, the lignin derived LPC suffers from lack of reproducibility and uniformity due to their structural heterogeneity. Paper-based precursors provide sustainable, cost-effective substrates for flexible electronics but require pretreatment with chemical (e.g., fire retardants) to limit carbonization and optimize LPC quality.

2.5. The Mechanism of Laser Processing Carbon

The laser reduction mechanism is widely used and discussed in the literature. The reduction mechanism is either based on the photochemical or photothermal effect, or a mixture of the dual effects. Smirnov et al. estimated the threshold energy (3.2 eV) for the photoreduction of GO.[86] According to them, the photochemical process can reduce GO with a laser of wavelength shorter than 390 nm, while the photothermal effect reduces GO with a laser of wavelength greater than 390 nm. However, recently, Fatkullin et al. discovered that under visible light, the laser-induced reduction of GO is predominantly governed by photochemical reactions, thereby disputing the long-standing assumption that photothermal effects are the principal driving force behind the process (**Figure 5a**).[54] They provided strong evidence that the governing mechanism in GO reduction in visible light is the photochemical cleavage of oxygen-containing groups. On another note, it has long been known that GO can be exothermally reduced within a temperature range of 200 to 230 °C by decomposing the oxygenated groups in the GO.[87,88] Laser irradiation of 248 nm, with 5 Hz frequency, and 200 mJ of energy on GO in solution can raise the temperature of the solvent by ≈40 °C, which is insufficient for exothermal reduction of GO. However, Huang et al. proposed that during laser irradiation, the GO flakes absorbed the laser energy and quickly converted it to heat. Although the temperature of the solvent cannot be raised above 50 °C, the temperature of GO flakes can be raised above 200 °C, which is sufficient for the photoreduction process.[24] According to Guo et al. femtosecond laser is capable of inducing both photochemical and photothermal.[89] The femtosecond laser irradiation causes an electronic excitation for the first few picoseconds, leads to a substantial weakening of the C—O electronic bonds close the upper valence band, causing rapid oxygen elimination process attributed to photochemical mechanisms. After adequate electron-hole recombination (≈>100 ps), the reduction process is controlled by the induced thermal effect. This interpretation was further validated by Arul et al., who employed various light sources-including nanosecond UV pulses at 248 nm, continuous-wave illumination at 788 nm, and 100 femtosecond near-infrared pulses at 800 nm for the reduction of GO.[52] GO reduction during the lasing process can be categorized into two parts; 1st photochemical reduction of oxygen, ablating the material, and second heat irradiation during the photochemical process rearranging the graphene-like domains thermally. These explanations also agree well with N. D. Orekhov, J. V. Bondareva, D. O. Potapov et al's. molecular dynamic simulations work.[73] They suggest that during the 1st laser pulse, rapidly gasify the oxygen functionalities on the GO surface and form radicals

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Table 1. The carbon sources, laser type, and its parameters, and corresponding applications.

Material/Carbon sources	Substrate	Laser type	Wavelength	Laser power	Other parameters	Application	Refs.
Graphene oxide film	CD/DVD media disc	Infrared laser	788 nm	5 mW	–	Electrochemical Capacitors	[26]
Graphene oxide	ITO glass	Continuous-wave lasers	(405, 532, And 633 nm)	from 10 μW to 2.35 mW	Scanning rate 50 mm min^{-1}	–	[54]
Cobalt/reduced graphene oxide	PET film	LightScribe DVD (HP-557 s) laser	783 nm	–	–	Micro supercapacitors	[55]
Hydrated GO and chloroauric acid ($HAuCl_4$)	Photo paper	femtosecond fiber laser	1030 nm	44 mW	Scanning rate 1 mm s^{-1}	Solid-state micro supercapacitors	[56]
Hydrated GO and chloroauric acid ($HAuCl_4$)	Photo paper	CW laser	532 nm	44 mW	Scanning rate 1 mm s^{-1}	Solid-state micro supercapacitors	[56]
Kapton PI	Polymer	CO_2 laser cutter system	10.6 μm	2.4 to 5.4 W with increments of 0.6 W.	Pulse duration ≈14 μs.	In-plane microsupercapacitors	[28]
Commercial PI polymer sheet		CO_2 infrared laser	10.6 μm	5 W	Pulse duration 14 μs	Microsupercapacitor	[61]
Boron-doped fluorinated polyimide (F-PI/BO_x)	Polyimide	Continuous semiconductor lasers	400–450 nm	3 W	–	Micro-supercapacitors	[63]
PSU (polysulfone), PES (polyethersulfone), and PPSU(polyphenylsulfone),	Polymer	CO_2 pulse laser	10.6 μm	50 W	Scan rate of 25%	Electrocatalytic hydrogen peroxide generation, and antimicrobial hybrid membrane	[31]
Alkaline lignin	–	CO_2 laser	10.6 μm	2.4% to 4.0% of the total laser power (75 W) with increment 0.4%.	Scan speed 75 in. s^{-1} Laser pulse 1000 dpi $in.^{-1}$.	Micro supercapacitors	[36]
Cellulose filter paper	Paper	CO_2 laser	10.6 μm	≈ 800 mW	Scan speed 30 mm s^{-1}	Mechanical sensing	[72]
Graphene oxide	Silicon	Fiber pulse laser (IR)	1.064 μm		Single laser pulse 200 ns ulse frequency 1.6 to 1000 kHz	–	[73]
Graphene oxide	–	Femtosecond laser pulse	800 nm	–	–	–	[74]
PEEK, PEI, and Kapton films	Polymer	CO_2 laser	10.6 μm	5% power (3.75 W),	5% speed (3.75 in/s),	Multifunctional Polymer Nanocomposites	[75]
Polyethersulfone (PES) (UP 010P)	Polyethersulfone (PES) (UP 010P)	CO_2 laser	10.6 μm	0.1% of the maximum power 50 W	Image density (1000 PPI), scan rate (25%)	Water purification membranes.	[76]
Teflon polytetrafluoroethylene	Teflon polytetrafluoroethylene	CO_2 laser	9.3 μm	≈5.0 W,	Image density 500 pulses $inch^{-1}$ scan rate 15 cm s^{-1}	Fabrication of Fluorinated Nanodiamonds or Fluorinated Graphene	[30]

(*Continued*)

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Table 1. (Continued)

Material/Carbon sources	Substrate	Laser type	Wavelength	Laser power	Other parameters	Application	Refs.
PDMS film	Poly(ethyleneterephthalate) (PET)	Diode laser	405 nm	800 mW	scanning speed (≈10 mm $^{-1}$s	Pressure sensor	[77]
Phenolic resin/Fe	PET film, glass slides, sticky notes and so on)	CO_2 laser	405 nm	500 mW	Scanning speed, 200 mm $^{-1}$s	Supercapacitors and electrochemical glucose sensors	[78]
Phenolic resin	–	Nd:YAG	1064 nm	–	One ms pulse duration	Supercapacitors	[79]
Phenolic resin	–	CO_2 laser	10.6 µm		Scan speed 30 mm/s, power density 2.4 kW/cm^2	–	[80]
MoS_2, metal oxide (e.g., CuO), noble metal-doped metal oxide (e.g., Ag/ZnO) and composite rmetal oxides (e.g., In_2O_3/Cr_2O_3). PI tape (Kapton)	PI tape (Kapton)	CO_2 laser	–	Power 18%	Speed 7%	Gas sensor	[81]
Nageia fleuryi leaves	Nageia fleuryi leaves	Ytterbium-doped fiber femtosecond laser	346 nm	350 mW	Speed (≈10 000 mm s^{-1})	Micro-supercapacitors	[82]
Phenolic paper	–	CO_2 laser	10.6 µm	≈2.1 W	Scan rate 8.5 mm s^{-1}, pulse durati ≈14 µs.	Electrochemical device	[83]
Phenolic resin-PVA	PR-PVA	CO_2 laser	10.6 µm	30 W	–	flexible electronics	[84]
Ni/sucrose mixture	Ni/sucrose mixture	CO_2 pulsed lase	10.6 µm	75 W	Writing speed of 304.8 cm/s (120 in./s)	–	[85]
			10.6 µ				

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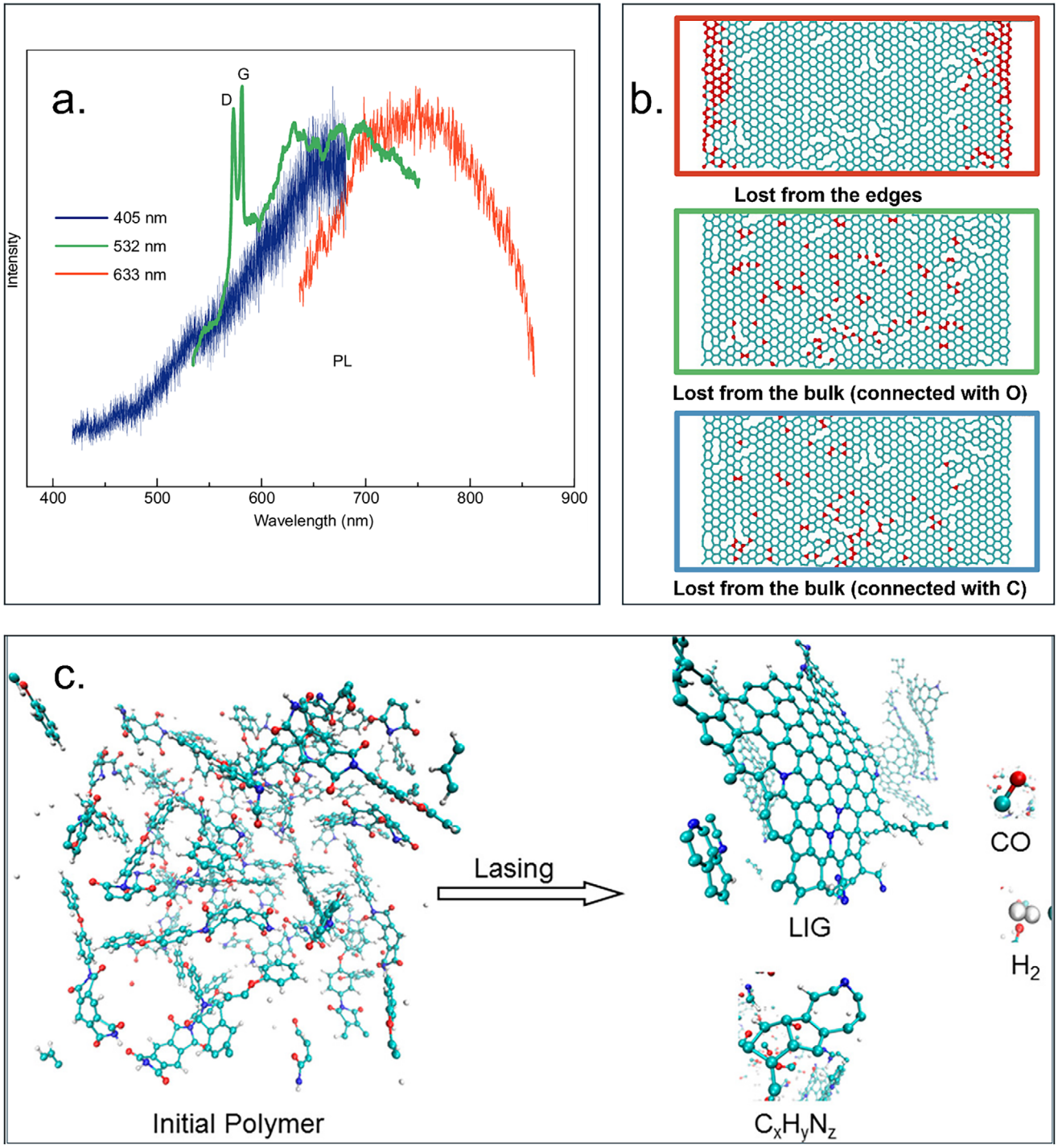


Figure 5. Laser reduction mechanism and molecular dynamic simulation results of the laser processed graphitic carbon formation a) Raman spectra of the graphene oxide film obtained under various excitation wavelengths at a constant power below 10 μW emphasize the D and G bands. Reproduced (Adapted) with permission.[54] Copyright 2024, Springer Nature. The carbon atoms initial positions within the GO sheet that eventually transition to the gas phase are marked with red spheres; b) atoms located at the edges of the GO sheet, bulk atoms initially bonded to oxygen-containing groups, and atoms from the bulk originally bonded only to other carbon atoms. Reproduced (Adapted) with permission.[73] Copyright 2022, Elsevier. Molecular dynamic simulation at a temperature ranging from 2500-3000 °C. c) Laser irradiation of polymers produces graphene, with the release of CO, H_2, and various hydrocarbon ($C_xH_yN_z$), Red spheres represent oxygen atoms, blue denote nitrogen, cyan indicates carbon, and white corresponds to hydrogen atoms. Reproduced (Adapted) with permission.[75] Copyright 2020, American Chemical Society.

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(-OH, -O, etc.). These radicals then initiate the oxidation process with atmospheric oxygen and form CO, CO_2, H_2O molecules (Figure 5b). Due to the high thermal conductivity of rGO/graphene, the material then undergoes rapid heating and cooling events during the next pulse, which is favourable for the rearrangement of covalent C—C bonds to increase structural order even further. However, according to Zhang and Miyamoto's findings from first-principles simulations of electron-ion dynamics, epoxy and hydroxyl functional groups present on GO sheets can be eliminated through femtosecond laser irradiation, while preserving the integrity of the graphene structure.[74]

In the case of polymer surfaces like PI (Kapton), poly(ether imide) (PEI), polybenzimidazole (PBM), polycarbonate (PC), the LIG formation using IR laser is more likely to be initiated by the photothermal process.[28,75] According to Lin et al., CO_2 laser irradiation on the PI surface causes lattice vibration, which raises the PI film's local temperature above 2500 °C.[28] The high local temperature causes the C═O, C—O, and N—C bonds to break. These atoms were then recombined and released as gaseous form (CO, CO_2, N_2, etc.). Aromatic rings in the PI sheet are then reorganized to form graphitic structures. Later, Tour et al. demonstrate that initial laser scribing converts the polymer substrate into amorphous carbon, and succeeding engraving converts the amorphous carbon into porous LIG.[37] The explanation was further verified by Vashisth et al's molecular dynamic simulation, suggesting that the best LPC can be formed at a temperature ranging from 2500–3000 °C (Figure 5c).[75] According to them, initially for a period of 0.2 ns an amorphous structure was formed which was then grown into ordered 5-, 6-, 7- 7-membered aromatic rings. They also suggest that laser reduction provides the best surface areas from PEEK and PEI, whereas PC may be the best choice for high electrical conductivity and PBM may form the best LIG due to the less oxygen in the polymer backbone.

The laser-processed sp^2 carbon offers several advantages compared to the other traditional synthesis routes, for example, it allows the user to customize the patterning without using any masks. Secondly, the released gas during the laser processing enables 3D interconnected porous and well-exfoliated graphene layers. Thirdly, laser processing engineers the atoms at the atomic level, i.e., the absorbed laser energy to the molecule simultaneously breaks and forms some chemical bonds. Heteroatomic doping seems a simple process to disturb atomic arrangement and tailor the electronic and electrochemical properties of LPC. Finally, the graphene sheets can easily be scratched out from the polymer substrate to employ it as a powder form or transferred to any arbitrary flexible and stretchable substrate.

Therefore, these possibilities create ample active sites inside the LPC for ions and electron transport, and the 3D hierarchical structure and pores enable volume expansion during the battery discharging and charging process. This kind of structural merits is beneficial for high energy/power density and long-term stability in rechargeable batteries. On the other hand, the molecular engineering of LPC via a heteroatom doping creates defects on the graphitic planes, which can endow unexpected electrochemical properties. Importantly, LPC electrodes are prepared on flexible *PI*, paper, Cu/Al foil which is proven to be good substrate to get rid of non-active binders and additives.

3. LPC-Based State-Of-The-Art Batteries

3.1. Electrode for Li-Ion Batteries

LIBs are now essential part of modern-day wearable/portable electronics, such as smartwatches, laptops, smartphones. With global rise of interest in electric vehicles (EVs) or hybrid electric vehicles (HEVs), the demand for LIBs, which can be incorporated into EVs/HEVs has increased significantly. In typical LIBs, the capacity output arises form Li ion intercalation to an active electrode material. Therefore, high diffusivity and conductive electrode is required to obtained high rate LIBs.[90,91] Graphite is the predominant anode material used in commercial lithium-ion batteries (LIBs), where lithium (Li) ions intercalate between its graphitic layers to form LiC_6, delivering a theoretical capacity of 372 mAh g^{-1}. However, the limited Li-ion diffusion within the bulk of graphite anodes restricts their performance in high-rate applications.[92] Thus, graphene-based electrodes have garnered attention due to their impressive electrical conductivity, great electrochemical surface area, and structural robustness.[93] Yoo and colleagues evaluated LIBs incorporating pristine graphene nanosheet anodes and reported specific capacities of ≈540 mAh g^{-1}, notably exceeding those of conventional graphite anodes.[94] Although the 1st cycle discharge capacity of LIB was great, the device suffered from poor coulombic efficiency (CE) due to the nonconductive polymer binders during the anode preparation.

Mukherjee et al. were the first to demonstrate the photothermal laser reduction of freestanding graphene oxide paper (**Figure 6a**) without the use of binders, resulting in graphene anodes for LIBs featuring a distinctive "open-pore" architecture.[95] They explain that laser heating reduces graphene oxide by the deoxygenation process.[26,96] The degassing process creates pores in microscale, defects, and voids in graphene paper, which enable better Li intercalation at high discharge/charge rates. At a high rate of 5 C a stable capacity of ≈370 mAh g^{-1} was reported and at a very high rate (40 C and 100 C) the laser-processed freestanding graphene electrode delivered an remarkable capacity of ≈156 and ≈100 mAh g^{-1}, respectively. Because of the open pore structure, the electrolyte penetrates deep into the laser-reduced graphene structure which reduces the Li-ion diffusion distance and results in a significant capacity retention at high discharge/charge rates.

Later on, Aslam et al. describe the synthesis of LIG foam on a flexible and foldable substrate. They used it directly as a self-supporting binder-less anode material for LIBs (Figure 6b).[97] In a typical method, the LIB electrode is prepared by mixing an active material with a polymer binder and carbon, and then the mixture is coated on a copper film.[98] Aslam et al. reported that LIGF can be used directly as an anode material with a folded design where the LIG foam was synthesized on a single side of flexible PI with two circles (3 mm diameter) with an electrical connection between them (inset of Figure 6b). One circle is used as the electrode and the other one was used a CC. Two types of surface morphology were observed in LIGF: rod-like structures at the top surface and overlapping sheet-like structures were found underneath it. However, they reported that the annealed LIGF shows better performance (280 μAh cm^{-2} @ 0.1 mA cm^{-2}) compared to as prepared LIGF (216 μAh cm^{-2}).

The laser processing method can make a carbon composite with other battery-type materials. Lu and colleagues narrated the

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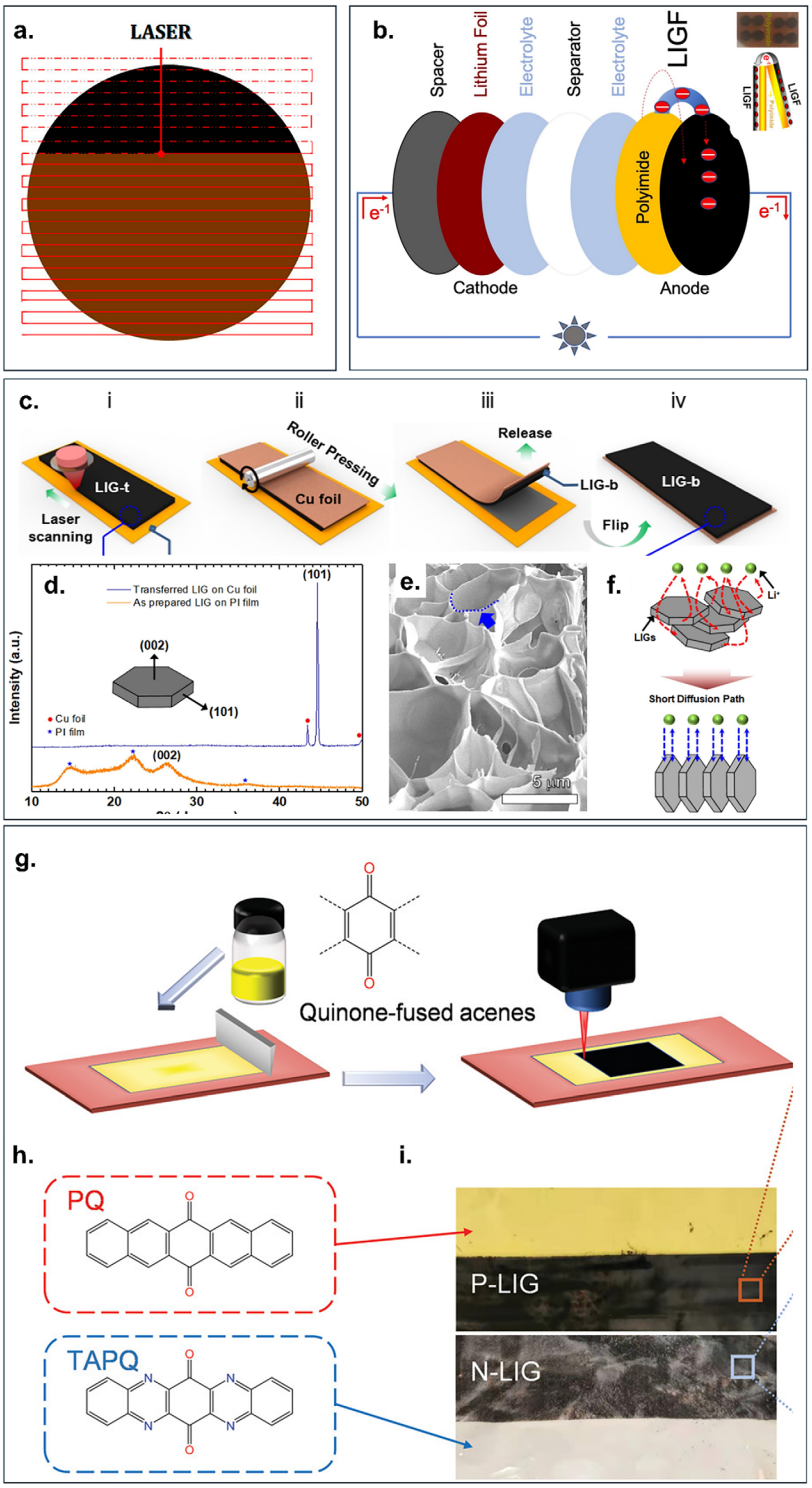


Figure 6. The concepts of LIB fabrication using different form of laser processed carbon. a) Schematic showing photothermal reduction of GO anode using a laser reduction system. Reproduced (Adapted) with permission.[95] Copyright 2012, American Chemical Society. b) coin cell assembly with folded design of LIG fiber on flexible *PI*. The inset showing the folded design schematic and the real electrodes. Reproduced (Adapted) with permission.[97] Copyright 2019, Elsevier. c) schematic diagram of LIG fabrication and transfer to a Cu substrate. d) XRD spectra of as-prepared LIG on flexible PI film (orange line) and transferred LIG on Cu substrate (blue line). e) Angled view of SEM micrograph of LIG. The blue dashed line corresponds to the (101) plane of LIG. f) Schematic showing the diffusion path of Li in closely stacked LIG and facet-oriented LIG.Reproduced (Adapted) with permission.[101] Copyright 2021, Elsevier. g) Schematic diagram of the fabricate the LIG films based on quinone-fused acenes onto the Cu substrate. h) Molecular assembly of 5,7,12,14-tetraaza-6,13-pentacenequinone (TAPQ) and 6,13-pentacenequinone (PQ), i) the digital image of the P-LIG transformed from the PQ (upper) and N-LIG from the TAPQ (bottom). Reproduced (Adapted) with permission.[103] Copyright 2022, Wiley-VCH.

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synthesis of SnO_2/LIG and SnO_2/N-doped LIG composite using a single-step lasing method. 1st, the precursor solution was prepared by mixing stannic oxide solution and a colloidal suspension of graphene oxide with urea as the source of nitrogen. Next, the precursor was coated on Cu foil and then dried in a vacuum oven, followed by the laser irradiation in Ar environment. They reported that the optimum laser fluence of 69.3 J cm^{-2} is required to obtain the SnO_2/LIG composite.[99,100] They reportedly achieved a reversible capacity of 830 mAh g^{-1} after 300 cycles at a current density of 300 mA g^{-1}. Even when the current density was increased tenfold to 3000 mA g^{-1}, the SnO_2/N-doped LIG composite electrode maintained a high reversible capacity of 600 mAh g^{-1}.The induced defect due to N doping promotes the Li ion insertion and diffusion rates in the LIG sheets.

In a similar way, Zhou et al. prepared MnO/Mn_3O_4/N-doped-LIG (MNLIG) hybrid anode for LIBs. The process includes the casting of precursor (Gelatine solutions containing manganese (II) phthalocyanine) onto copper foil, followed by laser treatment an ambient atmosphere.[44] The hybrid structure demonstrates outstanding rate capability, delivering 365 mAh g^{-1} at a current density of 2.0 A g^{-1}, along with a high reversible capacity of 992 mAh g^{-1} at 0.2 A g^{-1}. The electrochemical performance of the binder-free MNLIG anode surpasses that of the pristine NLIG anode. The unique structure of MNLIG framework, sufficient internal voids, and increased interlayer spacing were reported to be the reasons for short Li ion diffusion pathways.

Adopting nanostructured materials with high electrolyte accessible surface area and preferably structural alignment for favourable ion transport can also provide satisfactory rate kinetics in LIBs. By utilizing this concept, Shim et al. prepared facet-controlled holy graphene by transferring the 3D LIG onto a copper tape.[101] The LIG was prepared with the conventional CO_2 laser irradiation on a PI sheet, followed by a dry-release transfer of LIG on a copper CC (Figure 6c i–iv). The electrode delivers a high-rate characteristic of 114 mAh g^{-1} (at 20 A g^{-1}). The observed high density of (101) facets in the vertically standing LIG sheets was attributed to this high-rate performance (Figure 6d–f).

Next, Cho et al. modified the aluminum (Al) CC with laser-processed graphene to enhance the LIB performance.[102] A simple method of polyamic acid (PAA) deposition on the Al substrate, followed by curing the PAA into PI and photothermal conversion of PI into porous LIG using CO_2 laser was utilized to modify the Al CC. Incorporating a LIG layer on Al enhances the contact area between the active material and the electrode, thereby improving physical adhesion and lowering the interfacial resistance between the active layer and the electrode.

The LIBs performance was improved further by forming a graphene (P-LIG and N-LIG). Zhang et al. reported the transformation of small organic molecule pentacene quinone and tetraazapentacene quinone (TAPQ), into graphene (P-LIG and N-LIG) using a simple lasing method (Figure 6g–i).[103] The N-LIG anode for LIBs demonstrates a reversible capacity of 5863 mAh g^{-1} at a current density of 200 mA g^{-1} and maintains a high capacity exceeding 4000 mAh g^{-1} even under a significantly higher current density of 2000 mA g^{-1}. This is one of the best-reported capacity outputs of graphene-based LIB anode. Compared to polymer, the small organic molecule engineering using laser is reported to be much easier to develop high-performance LIB electrodes.

Very recently, Rahman et al presented a novel manufacturing method employing LIG on commercially available Kapton film to generate hexagonal pores.[68] These pores serve as vertical pathways for electron and ion movement during the processes of lithiation and delithiation, greatly improving the conductivity. The non-graphitized area of the Kapton film eliminates the need for binders, allowing it to function as a self-supporting electrode that enhances mechanical stability.

3.2. Current Collector for Li-Metal Battery

For high-end consumer electronics like electric vehicles/smartphones, high-energy-density batteries are essential. In this regard, Li metal has been considered a promising anode due to its outstanding theoretical capacity of 3860 mAh g^{-1}, negative redox potential of −3.040 V versus standard hydrogen electrode and ultra-low density of 0.53 g cm^{-3}.[104,105] But in practicality, the uncontrolled Li dendrite growth during continuous stripping plating results in a low CE, rapid capacity fading as well as safety hazards. The development of 3D porous materials as host structures for Li metal batteries has gained significant attention due to their ability to accommodate Li's volume changes during cycling and suppress dendrite formation. LIG can act as a 3D layer offering consistent nucleation sites, thereby reducing the ongoing side reaction related to the development of the solid electrolyte interphase (SEI) layer.

Yi et al. reported a stable Li metal anode by modulating Li nucleation and deposition kinetics with LIG.[39] They engineered a three-dimensional hierarchical LIG composite on a copper foil, which functioned as a CC. In this configuration, the porous LIG encapsulates a pillared framework composed of flexible polyimide. In this design, porous LIG envelops a pillared structure made of flexible polyimide. As illustrated in **Figure** 7a, the microstructure introduces a high density of defects, significantly lowering the Li nucleation barrier compared to untreated copper foil. Theoretical models indicate that defects serve as nucleation centers during heterogeneous Li deposition, allowing Li to nucleate without overpotential at current densities below 0.2 mA cm^{-2}. This leads to highly stable Li-metal anodes with Coulombic efficiencies around 99% (Figure 7b).

Similarly, Chen and colleagues reported a unique coating on Cu foil by multiple lasing using an infrared laser cutter on a commercial silicon adhesive tape.[106] The process produces a carbon-decorated SiO_x layer on Cu CC, which suppresses the formation of Li dendrites and delivers exceptional CE of 99.3% (2.0 mAh cm^{-2} at 2.0 mA cm^{-2}) and better cyclic stability as compared to uncoated Cu CC Figure 7c–e. According to the report, the coating material cannot be replicated with other controlled deposition methods like sputtering of SiO_x layer.

Recently, Xiao et al. presented and new design and synthesis of MnO_x nanoparticles anchored LIG a Li metal battery anode.[107] The process is similar to the previous studies of efficient CO_2 laser processing on a commercial PI sheet attached to a CC, copper foil. The structure consists of MnO_x@LIG on Cu CCs, flexible PI pillars, and MnO_x@LIG on and in between the pillars (Figure 7f). The flexible PI pillars function both as a binder and as a buffer to absorb Li's volume expansion during cycling. Additionally, the porous LIG structure combined with

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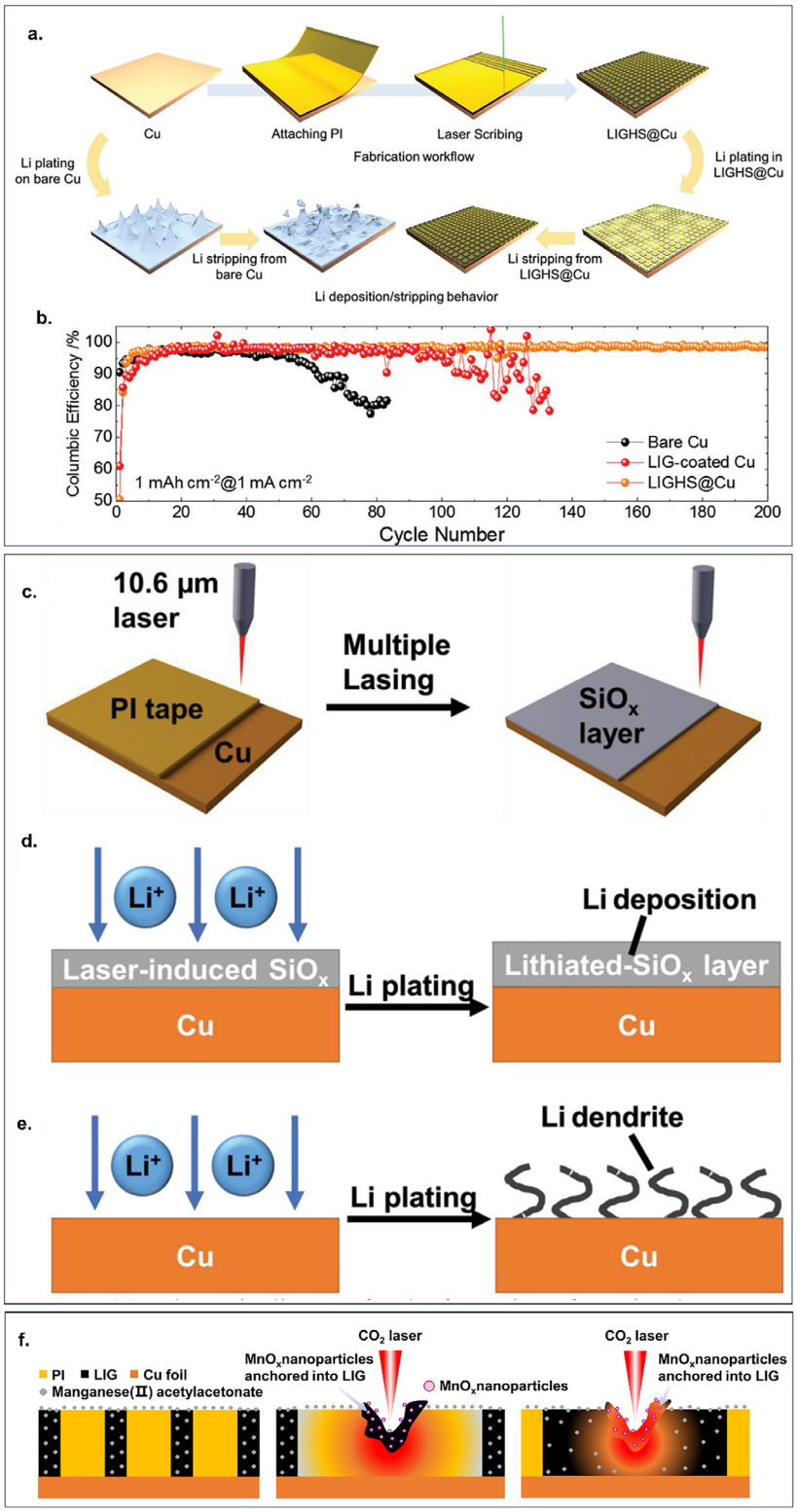


Figure 7. Laser fabrication strategies for Li-metal battery. a) schematic diagram of Cu, PI@Cu, and LIG hierarchical structure on Cu foil (LIGHS@Cu). b) Cyclic stability of Li-metal anode. The evolution of CE during Li deposition/stripping on bare Cu, on LIG-coated Cu, and in LIGHS@Cu at 1 mA cm^{-2}. Reproduced (Adapted) with permission.[39] Copyright 2019, Wiley-VCH. c) Schematic showing the development of Li-SiO_x layer on Cu CC by multiple lasing. d,e) Demonstration of the Li deposition on Cu substrate with a coating of Li-SiOx. Reproduced (Adapted) with permission.[106] Copyright 2020, Wiley-VCH. f) Schematic diagram showing the laser-based formation mechanism of the MnO_x@LIG. Reproduced (Adapted) with permission.[107] Copyright 2023, Elsevier.

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lithiophilic MnO_x reduces the Li nucleation overpotential, thereby facilitating dendrite-free Li deposition. The MnO_x@LIG electrode demonstrates exceptional rate capability, reaching up to 40 mA cm^{-2} with low nucleation overpotential. Even with a deposition of a very high areal capacity of 20 mAh cm^{-2}, the electrode surface maintained a smooth and dendrite-free morphology.

3.3. Laser-Processed Carbon for Li-S Battery

Beyond LIB, Li sulfur (Li-S) batteries have received extensive attention and are considered one of the leading successors amongst LIBs. The high theoretical prediction of specific capacity of ≈1675 mAh g^{-1} and outstanding energy density of ≈2600 Wh g^{-1} place Li-S batteries as one of the promising candidates as an alternative to conventional LIB.[108] Nevertheless, the shuttle effect due to the formation of soluble polysulfide induces instability inside the battery system, resulting in a low cyclic performance. The common strategy to mitigate the shuttle effect to engineer a carbon host that can confine sulfur.[109–112]

Li et al. demonstrated a pulsed UV laser processed sulfur and nitrogen-doped graphene electrode loaded with different nanoparticles like silver, platinum, silicon, and molybdenum sulfide.[113] This straightforward, single-step fabrication method yields a CC for Li-sulfur battery cathodes, supporting extended cycling with minimal capacity degradation. It was reported that after 500 cycles, the diffusion of polysulfides into the electrolyte was significantly suppressed in systems employing laser-produced doped graphene electrodes integrated with MoS_2 and cross-linked sulfur. The interaction of sulfur and oxygen-containing groups formed SO_2 species, and it played a significant role in recapturing the capacity and maintaining high stability. Here, the Li-S battery exhibited a low decay of charge capacity (0.002% per cycle) over 500 cycles, and surprisingly, charge capacity even increased over long cycles.

To mitigate the similar shuttling effect, Alshareef et al. prepared a freestanding 3D porous LSG interlayer that can host the nanosized carbon particles (**Figure 8a**). The freestanding LSG film minimized the polysulfide shuttling effect by both physisorption and chemisorption. The Li-S battery cell delivered an excellent capacity output of 1160 mAh g^{-1} at the rate of 0.25 C. The capacity retention was also impressive (80.4% after 100 discharge/charge cycles).[42]

3.4. Catalyst for Li-Air Batteries

The development of a bifunctional catalyst that is capable of both oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) is crucial to design a rechargeable metal-air battery. A key obstacle in the development of metal-air batteries is the significant overpotential and the slow kinetics of oxygen electrochemical reactions, specifically oxygen reduction during discharge and oxygen evolution during charging, occurring at the cathode surface.[114,115] To resolve these issues, the introduction of a carbon-based catalyst at the cathode surface is an easy and effective method.[116]

Tour et al. 1st represented a simple and low-cost method for the synthesis of a bifunctional ORR/OER catalyst by combining LIG and cobalt oxide (Co_3O_4) together (Figure 8b).[117] They proposed a single-step laser writing on a cobalt precursor-coated PI sheet, without using any liquid phase reaction or high-temperature pyrolysis. The catalyst was employed as the cathode material in both Li-oxygen (Li-O_2) and aqueous zinc-air battery systems. The resulting exhibits Co_3O_4 loaded LIG showed exceptional OER/ORR activity in alkaline electrolytes. The Li-O_2 battery exhibited outstanding cyclic stability of up to 242 cycles.

Furthermore, Tour et. al in 2019 used the concept that the ternary oxides of manganese, nickel, and iron exhibited a lower overpotential for OER and ORR, along with superior electron transfer numbers, outperforming the ternary oxides of cobalt, nickel, and iron to build a bifunctional catalyst combining Mn, Ni, Fe, and LIG. The research team prepared MnNiFe/LIG composites (M111/LIG and M311/LIG) with varying element ratios of Mn:Ni:Fe = 1:1:1 and Mn:Ni:Fe = 3:1:1 respectively.[116] Both catalysts exhibited excellent stability and prolonged life cycles in Li-air batteries, even without redox mediators like LiI, LiBr, tetrathiafulvalene (TTF), or TEMPO. The Li-air battery with M311/LIG catalyst achieved 350 stable cycles, while the M111/LIG catalyst enabled ≈300 cycles. The superior performance of M311/LIG was attributed to the discharge products' morphology which facilitates the charge transfer and ion diffusion. Another factor, the particle size of M311/LIG was smaller compared to M111/LIG, which represents a higher surface area and active sites. In addition, a higher Mn molar mass of M311/LIG speaks of more activity toward oxygen reduction.

In 2020 Ren et al. proposed a viable method for constructing a Li-O_2 battery by utilizing a dual polymer gel electrolyte (DPGE) in combination with a high-performance MnO_2/LIG cathode catalyst (Figure 8c).[43] The gel electrolyte structure provides high ionic conductivity, as ethylene oxide moieties in the gel electrolyte effectively solvate Li^+ ions. The DPGE demonstrated exceptional stability (2000 h) during the Li stripping/plating process without being short-circuited or increasing the interfacial resistance. Laser writing method was used for the catalyst production, which combined with DPGA electrolyte, enabling the fabrication of a practical Li-O_2 battery with a metallic Li anode. The MnO_2/LIG cathode catalyst was highly porous, electrically conductive, and chemically stable, which enables superior catalytic performance on ORR and OER reactions. The quasi-solid-state Li-O_2 battery demonstrated consistent discharge/charge cycling stability over 200 cycles with a cutoff capacity of 0.4 mAh cm^{-2}. Furthermore, it retained a high capacity of 2.0 mAh cm^{-2} for 50 cycles at a current density of 0.4 mA cm^{-2}, while maintaining the charging voltage below 4.5 V versus Li^+/Li. These finding demonstrates the applicability of LPC cathode catalysts for reversible Li-O_2 batteries and can be extended to other metal-air (Zn air) batteries as well.

3.5. Laser-Processed Carbon for Zn Ion and Zn Air Battery

Beyond LIB, zinc ion batteries (ZIB) have been extensively investigated due to the distinctive figure of merit of Zn anode. The low redox potential (−0.762 V vs RHE), outstanding theoretical capacity prediction of 820 mAh g^{-1}, and potential to commercialize due to large-scale availability in nature.[118–122] However, the uncontrolled formation of Zn dendrites during repeated Zn stripping/plating cycles, caused by uneven charge distribution on the

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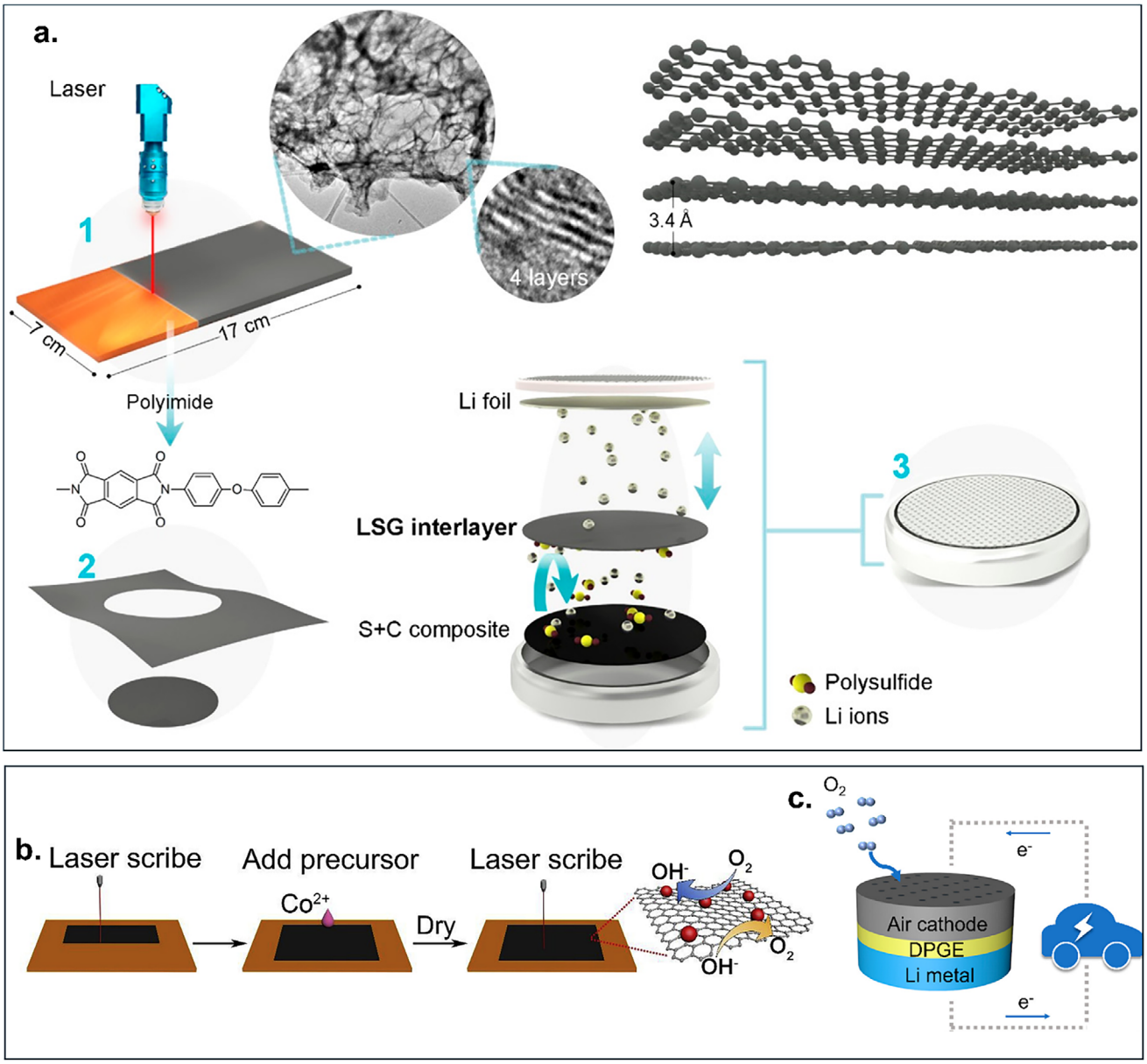


Figure 8. Li-S and Li-air battery fabrication strategy using various LPC. a) Graphic visual of the fabrication method of the free-standing LSGinterlayer and Li-S battery. Reproduced (Adapted) with permission.[42] Copyright 2020, American Chemical Society. b) Laser-based synthesis of Co_3O_4 particles decorated LIG catalyst. Reproduced (Adapted) with permission.[117] Copyright 2018, Elsevier. c) Schematic of Li-air battery with DPGE electrolyte. Reproduced (Adapted) with permission.[43] Copyright 2020, American Chemical Society.

Zn surface, can result in short circuits and ultimately lead to battery failure. Additionally, dendrites tend to break off from their base, forming "dead zinc", which contributes to capacity loss and reduces CE.[123–128]

In 2021, Alshareef et.al and his team prepared high surface area nitrogen-doped laser-processed graphene where evenly distributed nucleation sites are employed as an interlayer to regulate Zn nucleation and prevent the formation of Zn dendrites.[129] They used a simple method, where polyamic acid (PAA) containing 15 wt.% urea as a nitrogen source was blade-coated onto zinc substrate. After drying, the urea-infused *PI* film was directly converted into nitrogen-doped LSG (NLSG) through CO_2 laser writing (Figure 9a). The nitrogen functional groups in NLSG serve as zincophilic sites, exhibiting a robust binding affinity for Zn^{2+} ions, which induces uniform nucleation sites for Zn ions. NLSG exhibits superior electrochemical performance with a high CE of 99.4% and extended cycling stability of 200 h (Figure 9b). The NLSG-coated Zn//MnO_2 battery demonstrates superior performance compared to bare Zn anodes, maintaining a higher capacity retention of 72.3% versus 59.3% after 1200 cycles.

Later, Yang et al. used a similar concept of preparing laser-processed graphene to improve the nucleation. They presented a straightforward and efficient method for producing LIG, which was applied to the surface of Zn foil to create a composite

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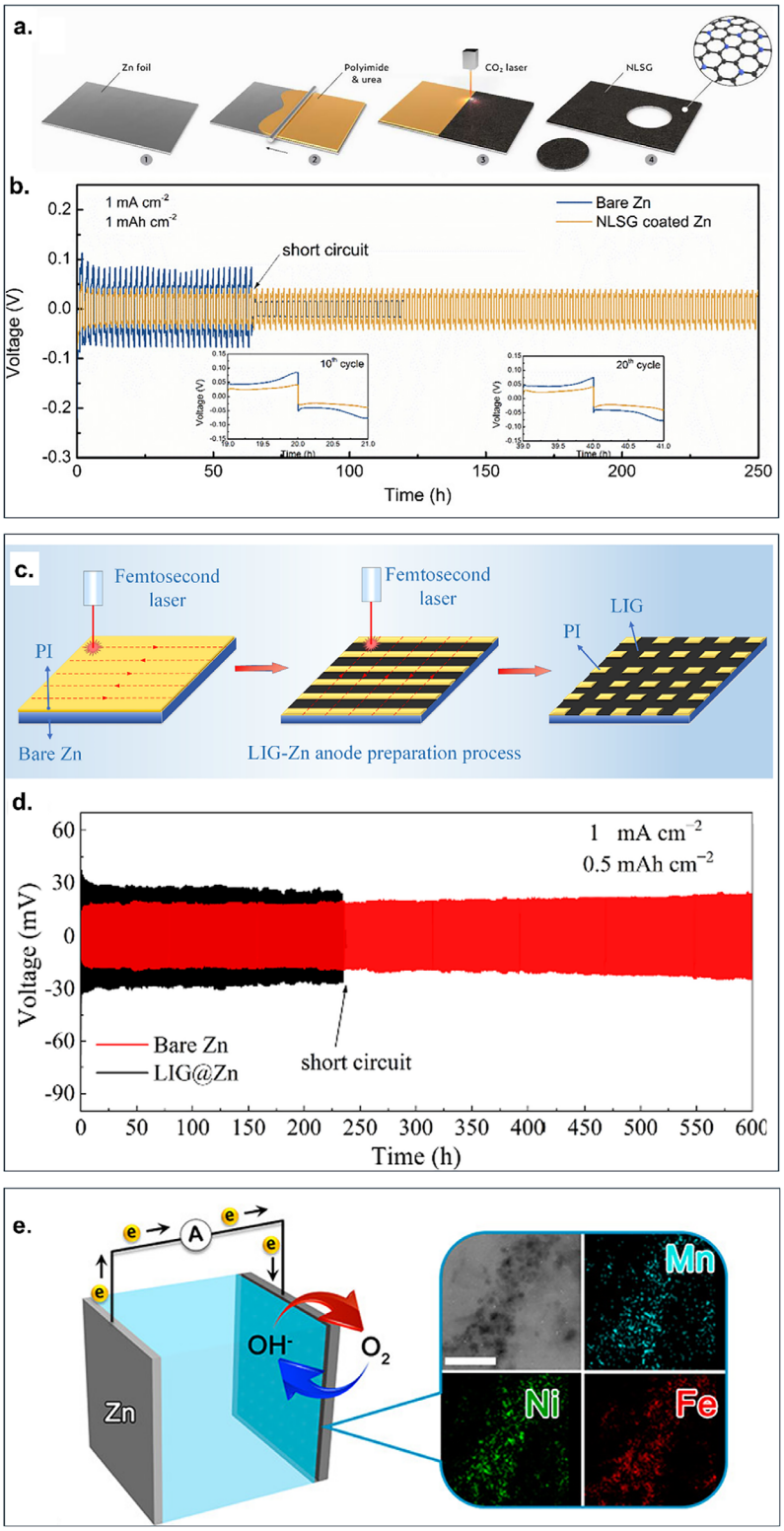


Figure 9. LPC structures for Zn ion and Zn air battery. a) Schematic illustration demonstrates the laser-based fabrication steps nitrogen-doped 3D graphene on Zn. b) Cyclic performance of symmetric cells of nitrogen-doped 3D graphene-modified Zn and bare Zn at 1 mA cm^{-2} with 1 mAh cm^{-2}. Reproduced (Adapted) with permission.[129] Copyright 2021, Wiley-VCH. c) Schematic of the femtosecond laser fabrication of LIG-Zn. d) Voltage profiles analyzed at 1 mA cm^{-2} with 0.5 mA h cm^{-2} capacity per cycle. Reproduced (Adapted) with permission.[130] Copyright 2023, Springer Nature. e) Schematic diagram showing the Zn air battery cell, the distribution of catalyst over laser processed carbon surface. Reproduced (Adapted) with permission.[135] Copyright 2019, American Chemical Society.

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structure (Figure 9c). This structure significantly lowers the nucleation overpotential of Zn and reduces Zn dendrite growth by enhancing Zn^{2+} nucleation behavior.[130] As compared to the bare Zn anode, the LIG on Zn demonstrates a much lower nucleation potential, resulting in uniform Zn ion deposition and minimizing dendrite formation. Zn-LIG anode exhibits a significantly longer cycle life, (>600 h) as compared to bare Zn anode (Figure 9d). Moreover, the LIG-Zn battery consistently retains an average coulombic efficiency exceeding 99.8% throughout more than 300 electroplating/annealing cycles, while the CE of bare Zn batteries drops significantly after just 80 cycles. The concave-convex pattern of LIG exhibits both hydrophilic and zincophilic properties, enabling full electrolyte wetting and ensuring uniform electroplating and dissolution across various currents and capacities.

On the other hand, Zn-air batteries (ZABs) are anticipated to play a crucial role in meeting the growing demand for high-performance energy storage and conversion systems. They offer several advantages, including a excellent theoretical prediction of energy density (≈1218 Wh kg^{-1} (6136 Wh L^{-1})), cost-effective, and enhanced safety.[131–134] Tour et al. introduced a simple laser-based method for synthesizing ternary metal oxide/LPC hybrid materials (Figure 9e), designed to serve as cathode catalysts for rechargeable ZABs.[135] They prepared four ternary metal oxide and LIG composite (LIG-CoNiFe-1/2 and LIG-MnNiFe-1/2) without using any wet chemistry or high-temperature pyrolysis. All the catalysts showed activity toward ORR and OER reaction in alkaline media, promoting their applicability in rechargeable ZABs. Among the four cathode catalysts, LIG-MnNiFe-1 emerged as the most efficient, achieving a peak power density of 98.9 mW cm^{-2}, a low overpotential of 0.88 V, an energy density surpassing 840 Wh kg^{-1}, and excellent cycling stability for over 500 cycles. The comprehensive design of porous carbon structure, which anchors the nanoparticles, and the optimized ratio of metal species to balance the OER/ORR activity were the reasons for the improved functioning of LIG-MnNiFe-1. Note that, LIG-MnNiFe-1 exhibited superior performance compared to the noble metal Pt/C+RuO_2 benchmark by delivering lower charge/discharge overpotentials, advanced power output, and improved reaction kinetics.

3.6. Laser-Processed Carbon for Na-Ion Battery

Sodium (Na) ion batteries (SIBs) are promising alternatives beyond LIBs due to their high abundance and low-cost resources of Na. Na ranked sixth according to its availability in the earth's crust (2.74%), which is 420 times compared to the Li (0.0065%).[46] As a result, the price of Na is nearly 100 times less than Li. Na exhibits similar electrochemistry to Li and possesses a low redox potential ($E^0_{(Na+/Na)}$ = −2.71 V versus the standard hydrogen electrode) and impressive theoretical predicted capacity of ≈1166 mAh g^{-1}.[136–138] But the fundamental difference between Na (1.02 Å) and Li (0.76 Å) is their ionic radius, which limits the Na ion intercalation in the commercial graphite structure, resulting in a low sodiation capacity of 35 mAh g^{-1}.[139,140] Therefore, the primary task to develop a high-performance SIB is to search for an appropriate framework of Na metal anode and develop a potential cathode material that can host Na ions conveniently.

In this direction, Zhang et al, reported a single-step laser writing process on urea-containing PI sheet-coated Cu CCs to fabricate high concentrations of nitrogen (≈13 at%) doped graphene anode for SIB (**Figure 10a**).[141] The versatile nature of the laser process ensures direct bonding between nitrogen-doped graphene and Cu CCs. They reported a reversible capacity output of 425 and 148 mAh g^{-1} at 0.1 A and 10 A g^{-1}, respectively. The expanded graphene structure (interlayer spacing ≈ 0.38 nm) due to nitrogen doping reduces the reaction resistance.

Later in 2023, Xiao et al. came up with the idea of a special gridded patterning of Sn NPs anchored LIG on Cu foil (named as Sn@LIG@Cu) to stabilize the Na metal anode (Figure 10b).[142] Sn nanoparticles participate in an in-situ alloying process to generate sodiophilic NaSn compounds, which lower the sodium nucleation overpotential and promote uniform, dendrite-free Na deposition. At the same time, the unique patterned structure and flexible PI columns serve as binders to buffer volume changes during plating/stripping. The Sn@LIG@Cu electrode demonstrates stable plating/stripping for 1300 cycles with a low overpotential of ≈ 5.2 mV and a high CE (>99.6%) at current density of 1 mA cm^{-2}. Moreover, it demonstrates outstanding rate capability at currents as high as 40 mA cm^{-2} and retains remarkable cycling stability over 500 h at 40 mA cm^{-2} and 20 mAh cm^{-2}, with 100% depth of discharge (DOD) and no evidence of dendrite formation.

In the same year, Alshareef et al, synthesized high oxygen-rich graphitic material from oil waste residue (fly ash carbon) just by simple laser irradiation (Figure 10c).[143] The laser technique reshapes the nanotexture of fly ash carbon from point-like to a planar structure while concurrently incorporating oxygenated groups that enhance Na adsorption and lower the energy barriers for Na^+ diffusion through graphitic layers. Compared to untreated fly ash carbon, the laser-treated anode delivers superior performance, including a higher reversible specific capacity (356 vs 119 mAh g^{-1}) at 0.1 A g^{-1}, a significantly improved initial CE (78% vs 41%), and a threefold increase in reversible capacity (250 vs 80 mAh g^{-1}) after 100 charge-discharge cycles.

Very recently, Richard B. Kaner and his team demonstrated a new synthetic strategy to construct vanadium oxide laser processed graphene (VOLG) material as a highly performing anode material for SIBs.[46] The two-step synthesis process involves 1st the laser-scribing on VCl_3 containing graphene oxide solution and subsequent calcination at 300 °C for 1 h. The VOLG anode, driven by Faradaic redox reactions and a mainly surface-controlled charge storage mechanism, exhibits exceptional fast-charging properties, achieving a high capacity output of 415 mAh g^{-1} at 0.2 A g^{-1} and 147 mAh g^{-1} at 20 A g^{-1} with excellent rate capability, while maintaining 87% cycle retention at 1 A g^{-1} after 1000 cycles. The battery performance of LPC based materials are summarized in **Table 2**.

4. Conclusions and Future Perspectives

Lasing methodology for microfabrication of carbon-based materials involves either pulsed or continuous wave lasing mode, with wavelengths varying from ultraviolet to visible to infrared. The basic principle lies in the absorption of laser energy, which causes either photothermal or photochemical conversion and crystallization of the precursor materials.[54] For carbon-based synthesis,

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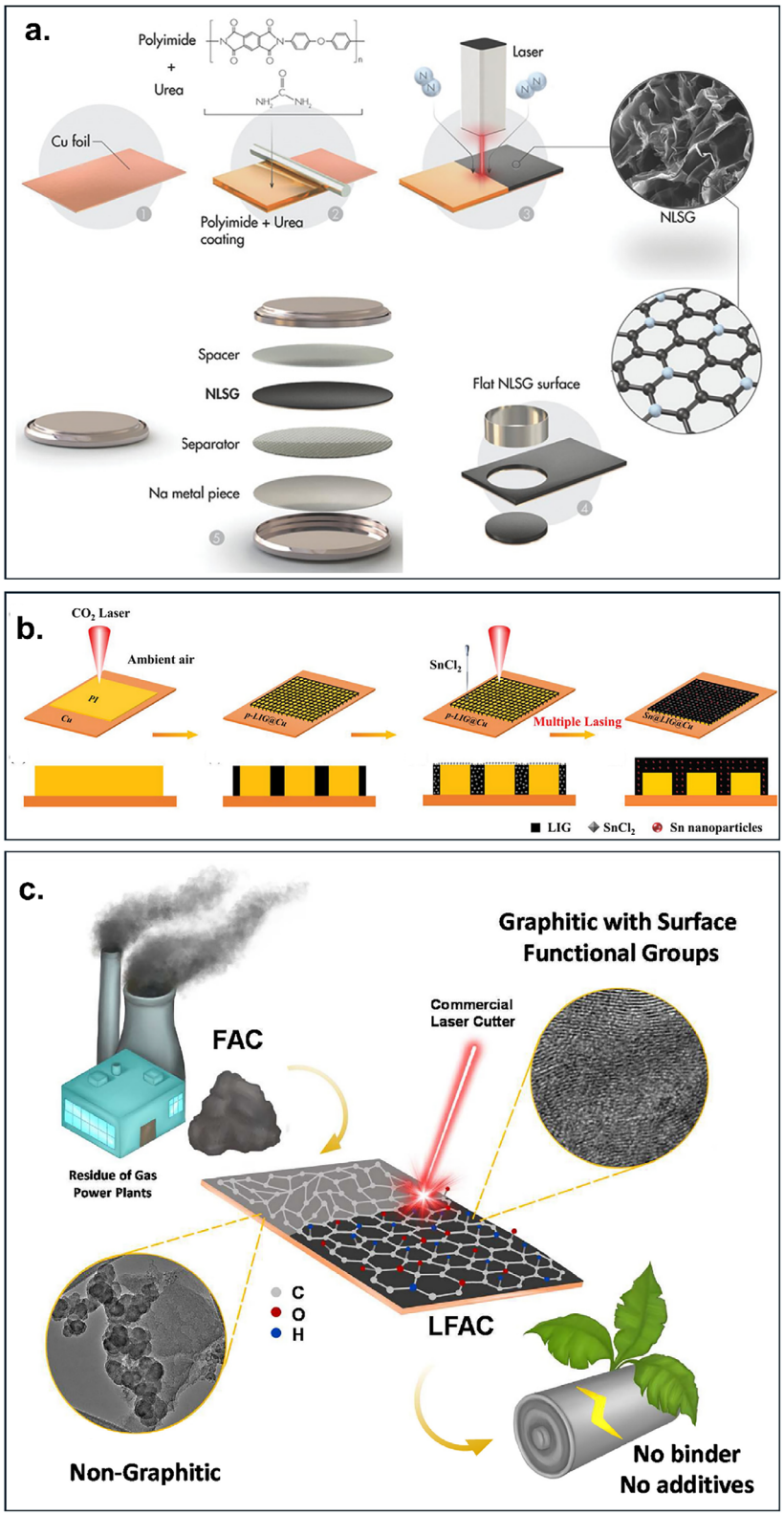


Figure 10. Laser fabrication strategy of various carbon for Na-ion battery. a) Schematic of the workflow showing the laser fabrication process of the nitrogen-doped 3D graphene on Cu foil and the assembly of the half-cell device. Reproduced (Adapted) with permission.[141] Copyright 2018, Wiley-VCH. b) The step-by-step fabrication of Sn@LIG@Cu and the side-view schematic of PI@Cu, p-LIG@Cu, $SnCl_2$-LIG@Cu, and Sn@LIG@Cu. Reproduced (Adapted) with permission.[142] Copyright 2023, Wiley-VCH. c) Fly ash upcycling process and turned into an anode advanced Na-ion battery. Reproduced (Adapted) with permission.[143] Copyright 2023, Elsevier.

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Table 2. The fabricated material, precursors, laser parameters, and their electrochemical performance.

Fabricated material	Precursor	Laser type	Laser power [W]	Wavelength	Electrochemical performance	Applications with advantages	Refs.
Reduced Graphene	Graphene oxide	CO_2 laser	≈4.8 W	9.3 µm	At 40 C and 100 C Capacity ≈156 mAh/g^{-1}, ≈100 mAh g^{-1}, 1000 cycles, power density ≈10 kW/kg^{-1}	Anode with high-power application for LIBs	[95]
LIG foam	Polyimide (PI)	CO_2 laser	30 W	≈10.6 µm	280 µAh cm^{-2} @ 0.1 mA cm^{-2}	Anode material with high energy density, lightweight for LIBs	[97]
SnO_2/N-doped graphene nanocomposite	$SnCl_2{\cdot}2H_2O$/ urea/graphene oxide/ Cu foil	Optical fibre laser	100 W	1064 nm	After 300 cycles capacity 830 mAh g^{-1}, @ current density 300 mA g^{-1} @ 3 A g^{-1}, reversible capacity 600 mAh g^{-1}	Anode material with improve Li storage capacities and rate performance Li storage	[99]
MNLIG composites	Manganese (II) phthalo-cyanine/gelatin/ Cu foils.	CO_2 laser	25 W	10.6 µm	Reversible capacity 992 mAh g^{-1} @ 0.2 A g^{-1} , excellent rate capacity 365 mAh g^{-1} @ 2.0 A g^{-1}, cycling stability 699 mAh g^{-1} after 400 cycles @ 0.2 A g^{-1}	Anode material with unique structure with abundant active sites, internal voids, and enhanced interlayer space for LIB	[44]
Three-dimensional holey graphene	Commercial PI films/copper tape	CO_2 laser	5.4 W	10.6 µm	Capacity ≈114 mAh g^{-1} at increased mass loading 3 mg cm^{-2}@ current density 20 A g^{-1}, 95% capacity charged within 3 min.	Unique structure, high-rate energy storage in LIB	[101]
Graphene-modified Al foils	Polyamic acid (PAA)/ Al	CO_2 laser	15 W	10.6 µm	Discharge capacity 115.7 mAh g^{-1} @ 1 C, capacity retention 98.5% after 500 cycles.	CC for LIB	[102]
Graphene (P-LIG and N-LIG)	Pentacene quinone and tetraazapentacene quinone (TAPQ),	CO_2 Laser	6 W	–	Reversible capacities of 5863 mAh g^{-1} @ 0.2 A g^{-1} and retains 1970 mAh g^{-1} @ 2 A g^{-1} after 500 cycles	Anode material with high-performance LIBs	[103]
LIG electrode	Kapton film	–	60 W	10.6 µm	Specific capacity of 581 mAh g^{-1} is maintained, with 99% coulombic efficiency @ 0.1C	Mass-producible, long-lasting, vertically conductive electrodes for LIBs	[68]
LIG (LIG)	Polyimide (PI) films/ copper foils	Laser engraving system	≈95 W cm^{-2}	532 nm	Coulombic efficiencies ≈99%.	Anode material with stability for Li-metal anode	[39]
Laser-induced silicon oxide (LI-SiO_x)	Commercial adhesive tape	CO_2 Laser	2% of 75 W	10.6 µm	Coulombic efficiency of 99.3% (2.0 mAh cm^{-2} at 2.0 mA cm^{-2})	Rechargeable Li metal anode	[106]
Li@MnO_x@LIG	MnO_x nanoparticle/PI tape	CO_2 laser, continuous wave	8.5% 50 W	10.6 µm	Rate capability 40 mA cm^{-2}, stably cycle for 3000 h with 100% depth of discharge@ 40 mA cm^{-2}	Scalable approach to stabilize anode Li metal anodes	[107]
Laser induced molybdenum sulphide	S and N doped graphene/various nanoparticles (silver, platinum, silicon, and molybdenum sulfide)	Pulsed UV laser	–	355 nm	At 0.5 C, the initial discharge capacity 768 $mAhg^{-1}$, with low fading rate 0.002% sustained 760 $mAhg^{-1}$ after 500 cycles	100% charge capacity Li sulphur (Li-S) batteries	[113]
Laser-scribed Graphene (LSG)	Commercial Kapton PI	Laser scribing	3.4 W	10.6 µm	Capacity of 1160 mAh g^{-1}, cycling stability 80.4% capacity retention after 100 cycles	Suppresses the polysulfide shuttling in Li-S batteries.	[42]

(*Continued*)

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Table 2. (Continued)

Fabricated material	Precursor	Laser type	Laser power [W]	Wavelength	Electrochemical performance	Applications with advantages	Refs.
Co_3O_4/LIG	$Co(NO_3)_2$ solution/ PI films	CO_2 laser	3% of full power	10.6 µm	ZAB power density of 84.2 mW cm^{-2} @ 100 mA cm^{-2}.	OER/ORR bifunctional catalyst, cathode material for metal-air batteries and aqueous ZABs and protic Li-O_2 battery.	[117]
MnO_2/laser-induced graphene (LIG)	MnSO4 Solution/ Kapton PI films	CO_2 laser	3% of full power	10.6 µm	Reversible capacity 2.0 mAh cm^{-2} @ current density 0.4 mA cm^{-2}	Cathode catalyst Li–O_2 batteries with high cyclable capacity	[43]
Nitrogen-doped 3D laser scribed graphene (NLSG)	PAA/urea film	CO_2 laser	4.5 W of 75 W	10.6 µm	Coulombic efficiency (99.4%) and cycling performance (200 h). NLSG-coated Zn//MnO_2 capacity retention (72.3% vs 59.3%) after 1200 cycles.	A large surface area combined with a uniformly distributed nucleation site interlayer helps regulate Zn nucleation behavior and inhibit the formation of Zn dendrites in Zinc-Ion Batteries.	[129]
Laser-induced graphene	Kapton tape (polyimide, PI) was attached to the Zn foil	Femto-second laser.	–	1064 nm	Initial nucleation overpotential 2.8 mV @ 1 mA cm^{-2}, cycle life reaches 600 h @ areal capacity 0.5 mAh cm^{-2}.	LIG-Zn anode Zn-ion batteries	[130]
Ternary metal oxide/graphene hybrid	Kapton PI films /1 M Co $(NO_3)_{2/}$ 1 M $MnSO_4$, 1 M $Ni(NO_3)_2$, and 1 M $Fe(NO_3)_3$ solutions	CO_2 laser	3% of full power	10.6 µm	LIG-MnNiFe- power density 98.9 mW cm^{-2}, I overpotential 0.88 V, energy density of >840 Wh kg^{-1}, cycling stability for >500 cycles	Cathode catalysts high active surface area with good electrical conductivity rechargeable ZABs	[135]
nitrogen-doped 3D graphene electrodes on Cu	Urea-containing polyimide solutions were cast onto copper (Cu) foils,	CO_2 laser	2.4 to 5.4 W	10.6 µm	Na-ion capacity up to 425 mAh g^{-1} @ 0.1 A g^{-1}, capacity 148 mAh g^{-1} @ current density 10 A g^{-1}	Anodes for high-rate Na-ion batteries	[141]
Sn nanoparticles-anchored laser-induced graphene on copper foil Sn@LIG@Cu)	PI tape copper foil/ stannous (II) chloride	CO_2 laser	8.5% of 50W	10.6 µm	Stably plate/strip for 1300 cycles with low overpotential (≈5.2 mV) and high Coulombic efficiency (CE > 99.6%) @ 1 mA cm^{-2}. rate performance up to 40 mA cm^{-2}	Scalable strategy to stabilize the sodium metal anodes	[142]
Fly ash carbon (FAC)	Fly ash carbon/copper sheet	CO_2 laser	6% of 75 W	10.6 µm	specific charge capacity 356 mAh/g @ 0.1 A/g,	Carbon anode of Sodium-ion batteries	[143]
Vanadium oxide laser-scribed graphene (VOLG)	GO powder /VCl_3 solution	CO_2 laser	12% of 40 W		Specific capacity 415 mA h g^{-1} @ 0.2 A g^{-1}, 147 mA h g^{-1} capacity retained @ 20 A g^{-1}	Anode material for sodium–ion batteries	[46]

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the photothermal effect predominates, and it requires a high-wavelength (10.6 µm) CO_2 laser. A recent study, challenging the widely held belief that photothermal effects are the main mechanism, reveals that photochemical reactions play a fundamental role in the laser-induced reduction of graphene oxide under visible light. However, the formation of the patterned surface is strongly dependent on the laser pulse width.

These laser-processed materials demonstrated promising potential for battery technology by acting as both anode/cathode or CCs. As an alternative of conventional mechanical exfoliation and chemical vapor deposition, the annealing treatment, laser-based approach offers advantages like energy efficiency, cost-effectiveness, and scalability. Notably, this approach enables the fabrication of binder-free and additive-free electrodes through maskless patterning under ambient conditions, significantly simplifying the manufacturing process; moreover, the precise control over electrode architecture and surface chemistry is essential for guiding uniform metal deposition, making it a critical step toward achieving dendrite-free next-generation batteries. The additional merit of hierarchical 3D porous structure, simplified doping methodology, and hybrid structure formation enables efficient ion intercalation/deintercalation with minimal volume change during the long cycling process. This review offers a fresh perspective on the integration of LPC nanostructures across diverse battery technologies, including Li-ion, Na-ion, Li-air, zinc-air, Li-metal, and Li-sulfur systems. While the process has already demonstrated significant strides in enhancing battery performance, it remains a relatively growing material with vast untapped potential. As research progresses, unlocking the full capabilities of laser processing approach could redefine the future landscape of high-performance, next-generation energy storage solutions. As the demand for safer, more efficient, and sustainable batteries continues to grow, the future of LPC in battery technologies looks increasingly promising.

Since its discovery in 2004, 2D graphene has earned significant interest worldwide, and over the last decade, there has been an intensive push to develop low-cost and green production methods for its large-scale synthesis and commercialization.[144] For example, the European Union's Graphene Flagship program, one billion-euro consortium comprising 150 partners, was created to bridge fundamental research and long-term commercialization across multiple sectors. To date, graphene has demonstrated promising applications wide range of areas such as energy storage, environmental remediation, and biomedicine. Nevertheless, despite the massive investments and intensive research, challenges persist for its widespread commercial adoption. Whereas direct laser writing can be performed under ambient atmosphere without any binders/additives or toxic chemicals. The ability to directly convert carbon-rich precursors (e.g., *PI*, wood, paper, or biomass) into conductive graphene-like structures using laser irradiation opens doors for eco-friendly mass production.[28,33,37,145] Future developments could focus on roll-to-roll laser scribing systems and integration with additive manufacturing, enabling large-scale, on-demand fabrication of carbon-based electrodes tailored for specific battery formats. The use of renewable or waste-derived precursors further aligns this technology with circular economy principles.[146–150]

Laser technique provides outstanding control over surface chemistry by allowing the heteroatom doping such as nitrogen, sulfur, oxygen, boron, silver, or phosphorus into the carbon lattice.[151] Among them, nitrogen is the most widely used dopant due to its similar atomic size to carbon. These doped LPC can be used to create single-atom catalysts (SACs), which present a transformative opportunity for advancing metal-air batteries, offering exceptional catalytic activity, atomic utilization, and tunable active sites.[152–155] The laser synthesis technique enables precise spatial control, defect engineering, and coordination modulation, creating high-performance SACs with minimal environmental footprint. These SACs enhance both oxygen reduction and evolution reactions, critical for rechargeable metal-air battery efficiency and longevity.[153–155] Looking forward, integrating laser-processed SACs into scalable architectures could overcome current limitations in durability, selectivity, and cost, paving the way for high-energy-density, dendrite-free, and eco-friendly next-generation metal-air batteries suited for grid-scale and wearable energy applications.

Laser-based synthesis and microfabrication techniques can be extended far beyond their currently explored applications, by adjusting the atmosphere during laser ablation, such as using nitrogen or NH_3 for nitrides, H_2S for sulfides, CH_4 for carbides, or even CO_2 for unique oxides and carbides diverse array of compound materials can be precisely engineered. While current limitations in laser nanomaterial processing persist in their resolutions, integrating advanced optical setups and hybrid systems could enable ultra-precise fabrication. Moreover, combining cold laser processing with thermal synthesis opens pathways to finely patterned, multifunctional electronics.[156] As laser technologies rapidly evolve, they will be poised to drive breakthroughs across a wide range of nanomaterial applications.

The conventional laser processing method relies on a line-by-line scanning method, taking multiple h to process even small areas, which limits its scalability for battery-related applications. An alternative and viable solution lies in the implementation of multiple laser beams, as suggested a review article by You et al.,[157] which can be efficiently integrated into roll-to-roll systems to greatly enhance production yield. Moreover, creating 3D nanostructures or patterned surfaces via conventional laser technique processing poses significant challenges; however, employing multiple parallel laser beams offers an effective solution to overcome both the structural and scalability restrictions. Note that the precision patterning is suitable for preparing micro-batteries and integration to compact electronics.[158,159]

The dendrite formation issue in conventional metal batteries (Li, Zn, Na) remains a major drawback to their commercial acceptance. LPC, with its high lithiophilic/zincophilic sites, offers an efficient platform to guide uniform metal deposition.[107,129,160] The research shows that laser-processed interlayers or host structures can significantly lower nucleation overpotential and accommodate metal plating/stripping over extended cycles. As battery technology moves toward solid-state and flexible configurations for applications in wearable electronics, the LPC electrodes can maintain performance under bending, stretching, and folding for the long term without dendrite formation.

Research in this battery technology field using LPC is still in its early stages, yet it progresses swiftly. There is enormous potential for the discovery of advanced anode/cathode, catalyst, and CCs with unique functionalities. The transition from lab-scale production to commercially viable technology is now within our

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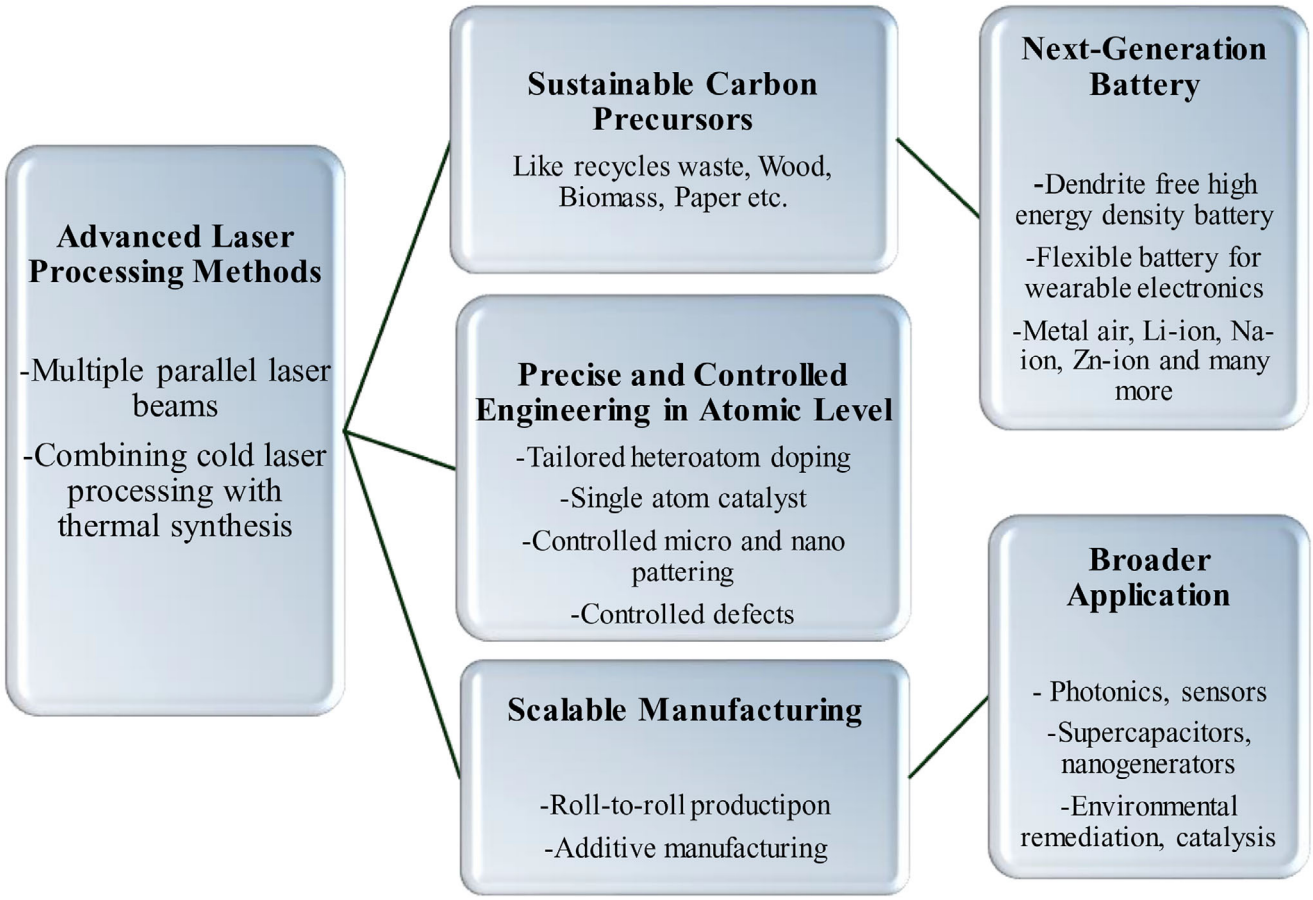


Figure 11. Future outlooks of laser processing technology and its applications in energy sector and beyond.

reach. A summary of future development directions in laser-based microfabrication of carbon structures with multidimensionality and functionalities for batteries and beyond represented as **Figure 11**. Continued advancement in laser technology is expected to play a crucial role in producing sustainable energy solutions and enabling more greener and efficient future.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

S.D. performed conceptualization, wrote, reviewed, and edited the manuscript, and also wrote the original draft; S.N. wrote, reviewed, and edited the draft; M.P. performed conceptualization, wrote, reviewed, and edited the manuscript.

Keywords

battery technology, laser processed carbon, laser writing, metal air battery, metal ion battery, metal-sulfur battery, sustainable battery materials

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